

THE DIVERGENCE THEOREM STOKES' THEOREM AND RELATED INTEGRAL THEOREM

THE DIVERGENCE THEOREM OF GAUSS:-

It states that if V is the volume bounded by a closed surface S and $\vec{A}$ is vector point function with continuous derivatives then

$$\iiint_V \vec{\nabla} \cdot \vec{A} \, dv = \iint_S \vec{A} \cdot \vec{n} \, ds = \oint_S \vec{A} \cdot d\vec{s}$$

where $\vec{n}$ is positive outward drawn unit normal to S .

STOKES' THEOREM:-

It states that if S is open, two-sided surface bounded by a simple closed curve C and if $\vec{A}$ has continuous derivatives then

$$\oint_C \vec{A} \cdot d\vec{r} = \iint_S (\vec{\nabla} \times \vec{A}) \cdot \vec{n} \, ds = \iint_S (\vec{\nabla} \times \vec{A}) \cdot d\vec{s}$$

where C is traversed in the positive direction as

$\vec{n}$ is positive outward drawn unit normal to S .

GREEN'S THEOREM IN THE PLANE:-

It states that if R is a closed region of the xy -plane bounded by a simple closed curve C and if M and N are continuous function of x and y having continuous derivatives in R . Then

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

where C is traversed in the positive direction.

Note: Green's Theorem is a special case of Stokes' Theorem. Also the

Gauss divergence Theorem is generalization of Green's Theorem because the Green's

Theorem for the ^{plane in which the} region R bounded by a ^{simple closed} curve C in plane while the Divergence Theorem for space in which the volume V is bounded by a closed surface S in space.

GREEN'S THEOREM IN THE PLANE:-

- ① Prove Green's Theorem in the plane if C is a closed curve which has the property that any straight line parallel to the coordinate axes cuts C in at most two points.

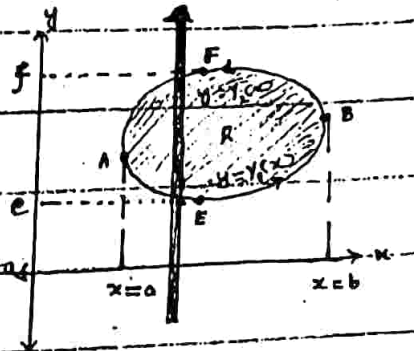
Sol:-

Let the equations of the curves

AEB and AFB are $y = Y_1(x)$ and

$y = Y_2(x)$ respectively.

if R is a closed region of xy -plane bounded by



simple closed curve C then

$$\iint_R \frac{\partial M}{\partial y} dx dy = \int_{x=a}^b \left[\int_{y=Y_1(x)}^{Y_2(x)} \frac{\partial M}{\partial y} dy \right] dx$$

$$= \int_{x=a}^b \left[M(x, Y_2(x)) - M(x, Y_1(x)) \right] dx$$

$$= \int_{x=a}^b [M(x, Y_2(x)) - M(x, Y_1(x))] dx$$

$$= \int_{x=a}^b M(x, Y_2(x)) dx - \int_{x=a}^b M(x, Y_1(x)) dx$$

$$= \int_{x=a}^b M(x, Y_2(x)) dx - \int_{x=a}^b M(x, Y_1(x)) dx$$

$$= \oint_C M dx$$

As,

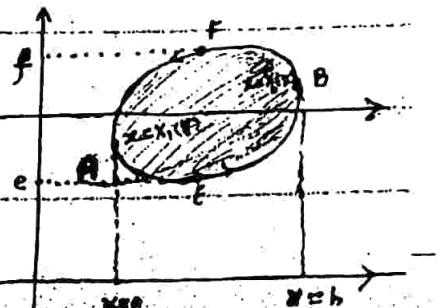
$$\oint_C M dx = - \iint_R \frac{\partial M}{\partial y} dx dy \rightarrow \textcircled{1}$$

Let the equations of the curves

EAF and EBF are $x = X_1(y)$ and

$x = X_2(y)$ respectively.

if R is a closed region of xy -plane bounded by



Simple closed curve: Green's Theorem

$$\iint_R \frac{\partial N}{\partial x} dx dy = \int_{y=e}^f \left[\int_{x=x_1(y)}^{x_2(y)} \frac{\partial N}{\partial x} dx \right] dy$$

$$= \int_{y=e}^f \left[N(x_2(y), y) - N(x_1(y), y) \right] dy$$

$$= \int_{y=e}^f [N(x_2(y), y) - N(x_1(y), y)] dy$$

$$= \int_{y=e}^f N(x_2(y), y) dy - \int_{y=e}^f N(x_1(y), y) dy$$

$$= \int_{y=e}^f N(x_2(y), y) dy + \int_{y=f}^e N(x_1(y), y) dy$$

$$= \oint_C N dy$$

As,

$$\oint_C N dy = \iint_R \frac{\partial N}{\partial x} dx dy \rightarrow (2)$$

Adding Equations (1) and (2)

$$\oint_C M dx + \oint_C N dy = - \iint_R \frac{\partial M}{\partial y} dx dy + \iint_R \frac{\partial N}{\partial x} dx dy$$

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

* * *

(2) Verify Green's Theorem in the plane for $\oint_C (xy + y^2) dx + x^2 dy$

where C is the closed curve of the region bounded

by $y=x$ and $y=x^2$

Sol:-

Given that

$$\oint_C M dx + N dy = \oint_C (xy + y^2) dx + x^2 dy \rightarrow (1)$$

Ans.

$$y = x : y = x^2$$

Sol.

$$x^2 = x$$

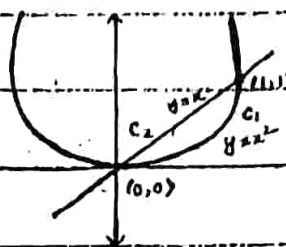
$$x^2 - x = 0$$

$$x(x-1) = 0$$

$$x = 0 : x = 1$$

$$\text{When } x = 0 \text{ Then } y = 0$$

$$\text{When } x = 1 \text{ Then } y = 1$$

Along $y = x^2$:-

$$\begin{aligned} \int_{C_1} M dx + N dy &= \int_{x=0}^1 (x \cdot x^2 + x^4) dx + x^2 d(x^2) \\ &= \int_{x=0}^1 (x^3 + x^4) dx + x^2 (2x) dx \\ &= \int_{x=0}^1 (x^3 + x^4 + 2x^3) dx \\ &= \int_{x=0}^1 (3x^3 + x^4) dx \\ &= \left(\frac{3}{4} x^4 + \frac{x^5}{5} \right) \Big|_0^1 \\ &= \frac{3}{4} + \frac{1}{5} \\ &= \frac{19}{20} \end{aligned}$$

Along $y = x$:-

$$\begin{aligned} \int_{C_2} M dx + N dy &= \int_{x=1}^0 (x \cdot x + x^2) dx + x^2 dx \\ &= \int_{x=1}^0 (x^2 + x^2 + x^2) dx \\ &= \int_{x=1}^0 (3x^2) dx \\ &= \left(x^3 \right) \Big|_1^0 \\ &= -1 \end{aligned}$$

The equation (1) becomes,

$$\begin{aligned} \oint_C M dx + N dy &= \int_{C_1} M dx + N dy + \int_{C_2} M dx + N dy \\ &= \frac{19}{20} - 1 = -\frac{1}{20} \quad \text{--- (2)} \end{aligned}$$

Since,

$$M(x, y) = xy + y^2 ; N(x, y) = x^2$$

$$\frac{\partial M}{\partial y} = x + 2y ; \frac{\partial N}{\partial x} = 2x$$

As,

$$\begin{aligned} \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy &= \iint_R (2x - x - 2y) dx dy \\ &= \iint_R (x - 2y) dx dy \\ &= \int_{x=0}^1 \int_{y=x^2}^x (x - 2y) dy dx \end{aligned}$$

$$= \int_{x=0}^1 \left[xy - y^2 \right]_{y=x^2}^x dx$$

$$= \int_{x=0}^1 (x^2 - x^2 - x^3 + x^4) dx$$

$$= \int_{x=0}^1 (-x^3 + x^4) dx$$

$$= \left(-\frac{x^4}{4} + \frac{x^5}{5} \right) \Big|_{x=0}^1$$

$$= -\frac{1}{4} + \frac{1}{5}$$

$$= -\frac{5+4}{20}$$

$$= -\frac{1}{20} \rightarrow (3)$$

From equation (2) and (3)

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

Hence Green's Theorem is verified.

(GREEN'S THEOREM FOR SIMPLY CONNECTED REGION)

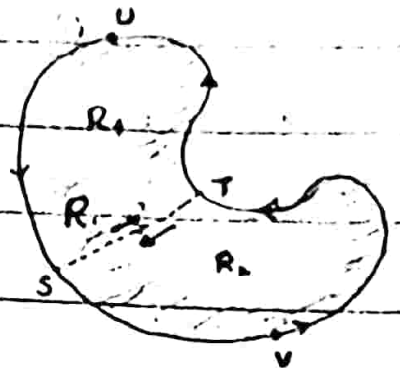
- (3) Extend the proof of Green's Theorem in the plane for the curves C for which lines parallel to the coordinate axes may cut C in more than two points.

Sol:-

Consider the closed curve C for which lines parallel

into the Cartesian axes may cut C in more than two points.

By constructing the line ST, the Region R is divided into two Regions R_1 and R_2



where R_1 and R_2 be Regions of simple closed

curves for which the Green's Theorem is applied.

As

$$\int_{STUS} Mdx + Ndy = \iint_{R_1} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy \rightarrow (1)$$

$$\int_{SVTS} Mdx + Ndy = \iint_{R_2} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy \rightarrow (2)$$

Adding Left Hand sides of equation (1) and (2)

$$\begin{aligned} & \int_{STUS} Mdx + Ndy + \int_{SVTS} Mdx + Ndy \\ &= \int_{ST} Mdx + Ndy + \int_{TUS} Mdx + Ndy + \int_{SVT} Mdx + Ndy + \int_{TS} Mdx + Ndy \\ &= \int_{ST} Mdx + Ndy + \int_{TUS} Mdx + Ndy + \int_{SVT} Mdx + Ndy + \int_{TS} Mdx + Ndy \\ &= \int_{TUS} Mdx + Ndy + \int_{SVT} Mdx + Ndy = \int_{ST} Mdx + Ndy \\ &= \int_{TUSVT} Mdx + Ndy = \int_C Mdx + Ndy \end{aligned}$$

Similarly By Adding Right Hand sides of Equation (1) and (2)

$$\iint_{R_1} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy + \iint_{R_2} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

Hence

$$\int_{TUSVT} Mdx + Ndy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

GREEN'S THEOREM IN VECTOR FORM (OR In Tangential Form)

④ Express Green's Theorem in the plane in vector notation.

Sol.

$$\text{Let } \vec{A} = M\hat{i} + N\hat{j}$$

$$\text{and } \vec{r} = x\hat{i} + y\hat{j}$$

$$d\vec{r} = dx\hat{i} + dy\hat{j}$$

$$\vec{A} \cdot d\vec{r} = (M\hat{i} + N\hat{j}) \cdot (dx\hat{i} + dy\hat{j})$$

$$\vec{A} \cdot d\vec{r} = Mdx + Ndy$$

$$\oint_C \vec{A} \cdot d\vec{r} = \oint_C Mdx + Ndy$$

GREEN'S THEOREM In Vector form or In Normal Form

$$\text{Let } \vec{A} = M\hat{i} + N\hat{j}$$

$$d\vec{r} = dx\hat{i} + dy\hat{j}$$

$$Mdx + Ndy = \vec{A} \cdot d\vec{r}$$

$$= \vec{A} \cdot d\vec{r} \text{ or}$$

$$= \vec{A} \cdot \vec{T} ds$$

where $\vec{T}$ is a unit Tangent vector to curve 'C'.

Let $\vec{n}$ is the outward drawn

Positive unit

Normal to 'C'

$$\text{As, } \vec{T} = \hat{k} \times \vec{n}$$

$$Mdx + Ndy = \vec{A} \cdot \vec{T} ds$$

$$= \vec{A} \cdot (\hat{k} \times \vec{n}) ds$$

$$= (\vec{A} \times \hat{k}) \cdot \vec{n} ds$$

$$= \vec{B} \cdot \vec{n} ds$$

where $\vec{B} = \vec{A} \times \hat{k}$

$$= N\hat{i} - M\hat{j}$$

$$\vec{B} \cdot \vec{n} = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}$$

The Green's Theorem in the plane becomes

$$\oint_C \vec{B} \cdot \vec{n} ds = \iint_R \vec{B} \cdot \vec{n} dR$$

$$\oint_C \vec{B} \cdot \vec{n} ds = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dR$$

$$\oint_C \vec{B} \cdot \vec{n} ds = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dR$$

Now

$$\vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ M & N & 0 \end{vmatrix}$$

$$= \left(-\frac{\partial N}{\partial z} \hat{i} - \frac{\partial M}{\partial z} \hat{j} + \hat{k} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \right)$$

$$= 0 - 0 + \hat{k} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right)$$

$$\vec{\nabla} \times \vec{A} = \hat{k} \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right)$$

$$(\vec{\nabla} \times \vec{A}) \cdot \hat{k} = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}$$

As Green's Theorem in the plane is

$$\oint_C Mdx + Ndy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

$$\oint_C \vec{A} \cdot d\vec{r} = \iint_R (\vec{\nabla} \times \vec{A}) \cdot \hat{k} dR \quad \text{where } dR = dx dy$$

which is the Green's Theorem in the plane in

Vector Notation (or in Tangential Form)

$$(\vec{\nabla} \times \vec{A}) \cdot \hat{k} = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}$$

$$\vec{\nabla} \cdot (\vec{A} \times \hat{k}) = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}$$

$$\vec{\nabla} \cdot \vec{B} = \frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}$$



⑥ Interpret physically the Green's Theorem in the plane in vector notation.

Sol:-

The Green's Theorem in the plane in vector notation is given by

$$\oint_C \vec{A} \cdot d\vec{r} = \iint_R (\vec{\nabla} \times \vec{A}) \cdot \hat{k} \, dx \, dy$$

If $\vec{A}$ denotes the force field acting on the particle. Then $\oint_C \vec{A} \cdot d\vec{r}$ is work done in moving the particle around the closed path C and determined by the value of $\vec{\nabla} \times \vec{A}$.

If $\vec{\nabla} \times \vec{A} = \vec{0}$ or $\vec{A} = \vec{\nabla} \phi$

Then the integral around a closed path is zero

This shows that the work done in moving a particle from one point to another is independent of the path followed or the force field is conservative.

Conversely,

Suppose that the integral around a closed path is zero

Then $\vec{\nabla} \times \vec{A} = \vec{0}$

This shows that in the plane, $\vec{\nabla} \times \vec{A} = \vec{0}$ is equivalent

to $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ if $\vec{A} = M\hat{i} + N\hat{j}$

(Hint Q#10, 11 Chapter 5)

⑥ Evaluate $\int_{(0,0)}^{(2,1)} (10x^4 - 2xy^3) dx - 3x^2y^2 dy$ along the path

$$x^4 - 6xy^3 = 4y^2$$

Sol:-

$$\text{As } \phi(x, y) = \int_{(0,0)}^{(2,1)} (10x^4 - 2xy^3) dx - (3x^2y^2) dy \rightarrow (1)$$

$$M(x, y) = 10x^4 - 2xy^3 \quad ; \quad N(x, y) = -3x^2y^2$$

$$\frac{\partial M}{\partial y} = -6xy^2 \quad ; \quad \frac{\partial N}{\partial x} = -6xy^2$$

$$\text{Since } \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

This shows that the integral is independent of the path followed.

Let us we choose a particular path the straight line segment from $(0,0)$ to $(2,0)$ and then from $(2,0)$ to $(2,1)$.

As

$$\begin{aligned} \phi(x, y) &= \int_{x=0}^2 10x^4 dx + \int_{y=0}^1 (-3x^2y^2) dy \\ &= \left[2x^5 \right]_0^2 - \left[4y^3 \right]_0^1 \\ &= 2(32) - 4(1)^3 \\ &= 64 - 4 \\ &= 60 \end{aligned}$$

Alternatively:-

$$\text{As } \phi(x, y) = \int_{(0,0)}^{(2,1)} (10x^4 - 2xy^3) dx - (3x^2y^2) dy \rightarrow (2)$$

$$M(x, y) = 10x^4 - 2xy^3 \quad ; \quad N(x, y) = -3x^2y^2$$

$$\frac{\partial M}{\partial y} = -6xy^2 \quad ; \quad \frac{\partial N}{\partial x} = -6xy^2$$

Since

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

This shows that the integral is independent of the

path followed. (or $\vec{\nabla} \times \vec{A} = \vec{0}$ if $\vec{A} = M\hat{i} + N\hat{j}$)

As $\phi(x, y)$ is an exact differential of $2x^5 - x^2y^3$

The equation (1) becomes

$$\phi(x, y) = \int_{(0,0)}^{(2,1)} d(2x^5 - x^2y^3)$$

$$= (2x^5 - x^2y^3) \Big|_{(0,0)}^{(2,1)}$$

$$= 2 \cdot 32 - 4 \cdot 1$$

$$= 64 - 4$$

$$= 60$$

* * *

Q. Show that the Area bounded by a simple closed curve C is given by $\frac{1}{2} \oint_C x dy - y dx$ ($\because$ Green's Theorem for Area formula)

Sol:—

The Green's Theorem in the Plane is,

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

Put $M = -y$ and $N = x$

So,

$$\oint_C -y dx + x dy = \iint_R \left(\frac{\partial}{\partial x}(x) - \frac{\partial}{\partial y}(-y) \right) dx dy$$

$$\oint_C x dy - y dx = \iint_R (1 - (-1)) dx dy$$

$$= 2 \iint_R dx dy$$

$$\oint_C x dy - y dx = 2A$$

$$A = \frac{1}{2} \oint_C x dy - y dx$$

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Q. Find the Area of the ellipse $x = a \cos \theta$, $y = b \sin \theta$

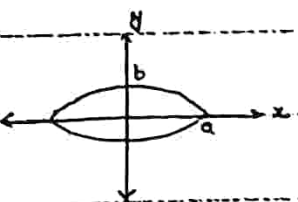
Sol: As,

$$\text{Area} = \frac{1}{2} \oint_C x dy - y dx \quad \text{--- (1)}$$

Given that $x = a \cos \theta$; $y = b \sin \theta$

$$dx = -a \sin \theta d\theta ; dy = b \cos \theta d\theta, \quad 0 \leq \theta \leq 2\pi$$

The equation (1) becomes



$$\text{Area} = \frac{1}{2} \int_0^{2\pi} (a \cos \omega)(b \sin \omega) - (b \sin \omega)(-a \sin \omega) d\omega$$

$$= \frac{1}{2} \int_0^{2\pi} (ab \cos^2 \omega + ab \sin^2 \omega) d\omega$$

$$= \frac{1}{2} \int_0^{2\pi} ab (\cos^2 \omega + \sin^2 \omega) d\omega$$

$$= \frac{ab}{2} \int_0^{2\pi} d\omega$$

$$= \frac{ab}{2} (2\pi)$$

$$= ab\pi$$

⑨ Evaluate $\oint_C (y - \sin x) dx + \cos x dy$ where C is the triangle of the adjoining figure.

① directly

② by using Green's Theorem in the plane.



Sol:-

① As,

$$\oint_C M dx + N dy = \oint_C (y - \sin x) dx + \cos x dy \quad \text{--- (1)}$$

Along C_1 :

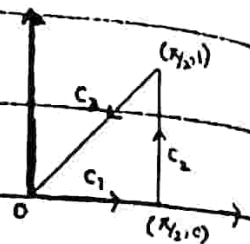
As $0 \leq x \leq \pi/2$, $y=0$, $dy=0$

The equation (1) becomes

$$\int_{C_1} M dx + N dy = \int_{x=0}^{\pi/2} (-\sin x) dx$$

$$= \cos x \Big|_0^{\pi/2}$$

$$= -1$$



Along C_2 :

As $0 \leq y \leq 1$, $x=\pi/2$, $dx=0$

The equation (1) becomes

$$\int_C Mdx + Ndy = \int_{y=0}^1 0 + 0 = 0$$

Along C_2 :-

The equation of straight line $\overline{BO}$ is

$$\frac{x-0}{\pi/2-0} = \frac{y-0}{1-0}$$

$$\frac{2x}{\pi} = y$$

$$\frac{2}{\pi} dx = dy$$

And $0 \leq x \leq \pi/2$

The equation (1) becomes

$$\int_{C_2} Mdx + Ndy = - \int_{x=0}^{\pi/2} \left(\frac{2x}{\pi} - \sin x \right) dx + \cos x \cdot \frac{2}{\pi} dx$$

$$= - \left(\frac{x^2}{\pi} + \cos x + \frac{2}{\pi} \sin x \right) \Big|_0^{\pi/2}$$

$$= - \left(\frac{\pi^2}{4} + 0 - \frac{2}{\pi} - 0 + 1 - 0 \right)$$

$$= - \frac{\pi}{4} - \frac{2}{\pi} + 1$$

Hence

$$\oint_C Mdx + Ndy = \oint_{C_1} Mdx + Ndy + \oint_{C_2} Mdx + Ndy + \oint_{C_3} Mdx + Ndy$$

$$= -\pi - \frac{\pi}{4} - \frac{2}{\pi} + 1$$

$$= -\frac{\pi}{4} - \frac{2}{\pi}$$

(b) As,

$$\oint_C Mdx + Ndy = \oint_C (y - \sin x) dx + \cos x dy$$

where $M(x, y) = y - \sin x$; $N(x, y) = \cos x$

$$\frac{\partial M}{\partial y} = 1 ; \quad \frac{\partial N}{\partial x} = -\sin x$$

The equation of straight line is

$$\frac{x-0}{\pi/2-0} = \frac{y-0}{1-0}$$

$$\frac{2x}{\pi} = y$$

As $0 < x < \pi$, $0 \leq y \leq \frac{2x}{\pi}$

The Green's Theorem in the plane is

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

$$= \int_{x=0}^{\pi/2} \int_{y=0}^{\frac{\pi-x}{\pi}} (-\sin x - 1) dy dx$$

$$= \int_{x=0}^{\pi/2} \left[-y \sin x - y \right]_0^{\frac{\pi-x}{\pi}} dx$$

$$= \int_{x=0}^{\pi/2} \left[-\frac{2x}{\pi} \sin x - \frac{2x}{\pi} \right] dx$$

$$= -\frac{2}{\pi} \int_{x=0}^{\pi/2} (x \sin x + x) dx$$

$$= -\frac{2}{\pi} \left[-x \cos x + \sin x + \frac{x^2}{2} \right]_0^{\pi/2}$$

$$= -\frac{2}{\pi} \left[0 + 1 + \frac{\pi^2/4}{2} + 0 - 0 - 0 \right]$$

$$= -\frac{2}{\pi} \left[1 + \frac{\pi^2}{8} \right]$$

$$= -\frac{2}{\pi} - \frac{\pi}{4}$$

GREEN'S THEOREM FOR MULTIPLY-CONNECTED REGION

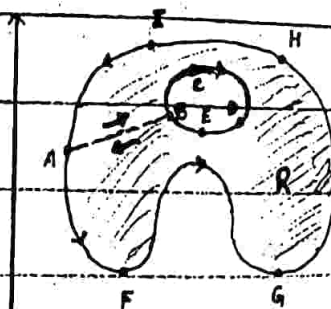
(10) Show that Green's Theorem in the plane is also valid for a multiply-connected Region R.

Sol:-

Let R be the Doubly Connected Region.

Constructing a Cross cut AB

which connecting the exterior and interior boundaries.



The Region bounded by ABCDEFGH is simply connected for which the Green's Theorem is valid.

$$\oint_{ACDEBFGHI} Mdx + Ndy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

$$\oint_{ACDEBFGHI} Mdx + Ndy = \int_{AB} + \int_{BCDEB} + \int_{BA} + \int_{AFGHIA}$$

$$= \cancel{\int_{AB}} + \int_{BCDEB} - \cancel{\int_{AB}} + \int_{AFGHIA}$$

$$= \int_{BCDEB} + \int_{AFGHIA}$$

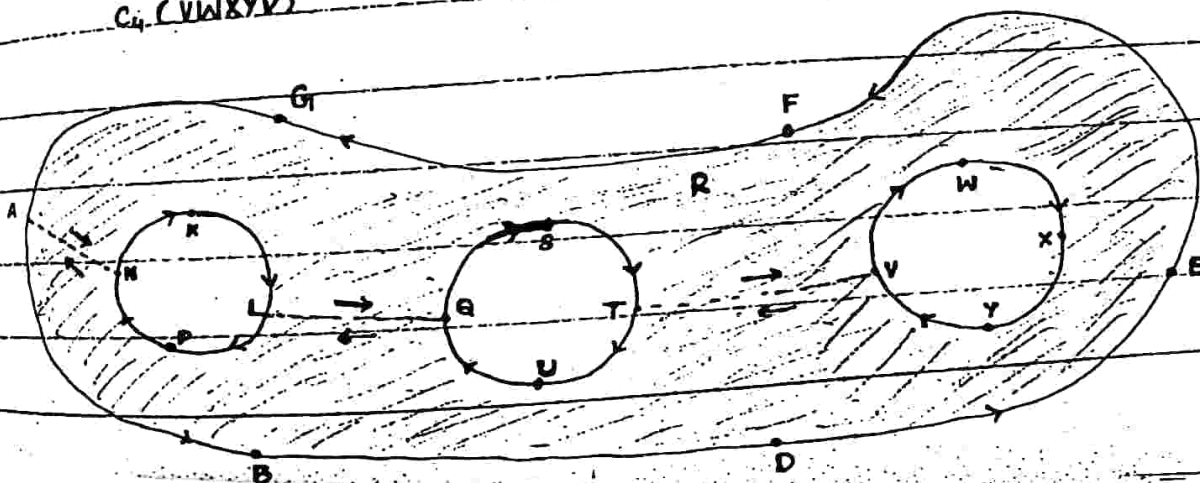
let C_1 be the curve of $BCDEB$ and C_2 be the curve of $AFGHIA$ and C is the boundary of R then

$$\oint_{ACDEBFGHI} Mdx + Ndy = \int_{C_1} + \int_{C_2}$$

$$= \oint_C$$

$$\oint_C Mdx + Ndy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

(11) Show that Green's Theorem in the plane holds for the region R of the figure below, bounded by the simple closed curves C_1 (ABDEFGH), C_2 (HKLPH), C_3 (QSTUQ) and C_4 (VWXYZV).



Solr

Let "R" be the multiply connected Region.

Constructing the cross cuts AH, LQ and TV which

Connecting the exterior and interior boundary.

The Region bounded by AHKLQSTUVWXYV TUQLPHABDEFGA is

Simply Connected for which Green's Theorem is valid.

$$\oint_{\text{AHKLQSTUVWXYV TUQLPHABDEFGA}} Mdx + Ndy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

As

$$\oint_{\text{AHKLQSTUVWXYV TUQLPHABDEFGA}} Mdx + Ndy = \int_{\text{AH}} + \int_{\text{HKL}} + \int_{\text{LQ}} + \int_{\text{QST}} + \int_{\text{TV}} + \int_{\text{VWXYV}} + \int_{\text{VT}} \\ + \int_{\text{TUQ}} + \int_{\text{QL}} + \int_{\text{LPH}} + \int_{\text{HA}} + \int_{\text{ABDEFGA}}$$

$$= \int_{\text{AH}} + \int_{\text{HKL}} + \cancel{\int_{\text{LQ}}} + \int_{\text{QST}} + \cancel{\int_{\text{TV}}} + \int_{\text{VWXYV}} + \cancel{\int_{\text{VT}}}$$

$$+ \int_{\text{TUQ}} - \cancel{\int_{\text{LQ}}} + \int_{\text{LPH}} - \cancel{\int_{\text{HA}}} + \int_{\text{ABDEFGA}}$$

$$= \int_{\text{HKL}} + \int_{\text{QST}} + \int_{\text{VWXYV}} + \int_{\text{TUQ}} + \int_{\text{LPH}} + \int_{\text{ABDEFGA}}$$

$$= \left(\int_{\text{HKL}} + \int_{\text{LPH}} \right) + \left(\int_{\text{QST}} + \int_{\text{TUQ}} \right) + \int_{\text{VWXYV}} + \int_{\text{ABDEFGA}}$$

$$= \int_{\text{HKLPH}} + \int_{\text{QSTUQ}} + \int_{\text{VWXYV}} + \int_{\text{ABDEFGA}}$$

$$= \int_{C_1} + \int_{C_2} + \int_{C_3} + \int_{C_4}$$

$$= \int_C \quad (\text{where } C \text{ is the boundary of } R)$$

$$\text{So, } \oint_C Mdx + Ndy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

Multiply Connected Region $\Rightarrow$ Doubly Connected Region

Q2) Prove that $\oint_C Mdx + Ndy = 0$ around every closed curve C in a simply connected region iff $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ everywhere in the region.

Sol-

Suppose that $\oint_C Mdx + Ndy = 0$ around every closed curve C in simply connected region.

Now we are to show that $\frac{\partial N}{\partial x} = \frac{\partial M}{\partial y}$ everywhere in the region.

We contrary supposed that

$$\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} > 0 \text{ at point } P$$

Then from the continuity of derivatives, $\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} > 0$ in some region A surrounding P .

Region A of P
is some neighborhood
of P

C is the boundary of A then by Green's Theorem in the plane

$$\oint_C Mdx + Ndy = \iint_A \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy > 0$$

This is the contradiction so our supposition is wrong.

Similarly $\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} < 0$ is not possible.

Hence $\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 0$ at all points.

Conversely,

Suppose $\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} = 0$ everywhere in the region.

Let R be a closed region of xy -plane bounded

by a simple closed curve C and M and N are

The continuous function of x and y have continuous derivatives in R . Then by Green's Theorem in the plane

$$\oint_C Mdx + Ndy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

As,

$$\oint_C Mdx + Ndy = 0$$

Alternatively,

For simple closed curve C

$$\oint_C \vec{A} \cdot d\vec{r} = \oint_C Mdx + Ndy$$

where $\vec{A} = M\hat{i} + N\hat{j}$

and

$$\nabla \times \vec{A} = \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) \hat{k}$$

we know

$$\oint_C \vec{A} \cdot d\vec{r} = 0 \quad \text{iff} \quad \nabla \times \vec{A} = \vec{0}$$

$$\text{Therefore } \oint_C Mdx + Ndy = 0 \quad \text{iff} \quad \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

(13) Let $\vec{F} = \frac{-y\hat{i} + x\hat{j}}{x^2 + y^2}$

(a) calculate $\nabla \times \vec{F}$

(b) Evaluate $\oint \vec{F} \cdot d\vec{r}$ around any closed path.

and Explain the Results

Sol.

(a) As, $\vec{F} = \frac{-y\hat{i} + x\hat{j}}{x^2 + y^2}$

$$= \frac{-y}{x^2 + y^2} \hat{i} + \frac{x}{x^2 + y^2} \hat{j}$$

Now

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{-y}{x^2 + y^2} & \frac{x}{x^2 + y^2} & 0 \end{vmatrix}$$

$$= \hat{i} (0) + \hat{j} (0) + \hat{k} \left(\frac{y^2 - x^2}{(x^2 + y^2)^2} - \frac{y^2 - x^2}{(x^2 + y^2)^2} \right)$$

$$= \hat{i} (0) + \hat{j} (0) + \hat{k} (0)$$

$$= \vec{0} \quad (\text{in any Region Excluding } (0,0))$$

As $\oint_C \vec{F} \cdot d\vec{r} = \oint \left[\frac{-y\hat{i} + x\hat{j}}{x^2 + y^2} \right] \cdot [dx\hat{i} + dy\hat{j}]$

$$= \oint \frac{-y dx + x dy}{x^2 + y^2} \rightarrow (1)$$

Let $x = r \cos \theta$; $y = r \sin \theta$

$dx = -r \sin \theta d\theta$; $dy = r \cos \theta d\theta$

The Equation (1) Becomes

$$\begin{aligned} \oint_C \vec{F} \cdot d\vec{r} &= \oint -r \sin \theta \cdot -r \sin \theta d\theta + r \cos \theta \cdot r \cos \theta d\theta \\ &= \oint \frac{r^2 \cos^2 \theta + r^2 \sin^2 \theta}{r^2 (\sin^2 \theta + \cos^2 \theta)} d\theta \\ &= \oint d\theta \rightarrow (2) \end{aligned}$$

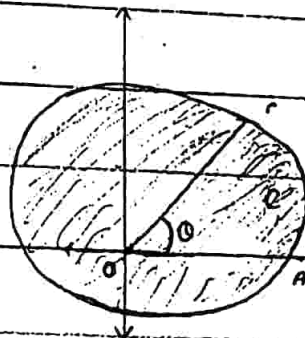
For a closed curve C in which the origin is including:-

As $\theta = 0$ at A and $\theta = 2\pi$ after a complete rotation at A.

so $0 \leq \theta \leq 2\pi$

The equation (2) becomes

$$\begin{aligned} \oint_C \vec{F} \cdot d\vec{r} &= \int_{\theta=0}^{2\pi} d\theta \\ &= \theta \Big|_0^{2\pi} \\ &= 2\pi \end{aligned}$$



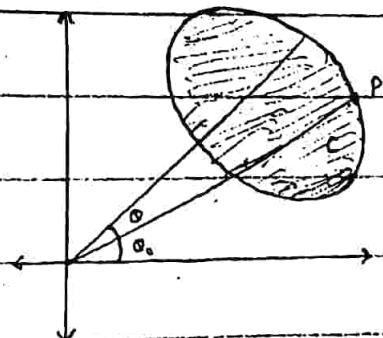
For a closed curve C in which the origin is excluding:-

As $\theta = \theta_0$ at P and $\theta = \theta_0$ after a complete rotation at P

so $\theta_0 \leq \theta \leq \theta_0$

The equation (2) becomes

$$\begin{aligned} \oint_C \vec{F} \cdot d\vec{r} &= \int_{\theta=\theta_0}^{\theta_0} d\theta \\ &= \theta \Big|_{\theta=\theta_0}^{\theta_0} \\ &= 0 \end{aligned}$$



Note: Comparison with Q-N thm

where $\nabla \times \vec{F} = \vec{0}$ so, $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ Then integral should be independent of path

formed but here M and N have continuous derivatives in any region including and

THE DIVERGENCE THEOREM,

(17) (a) Express the divergence Theorem in words

and

(b) write it in Rectangular form.

Sol:-

(a) As,

$$\iiint_V \nabla \cdot \vec{A} \, dv = \iint_S \vec{A} \cdot \vec{n} \, ds = \oint_S \vec{A} \cdot d\vec{s}$$

The Surface integral of the Normal component of a Vector field $\vec{A}$ taken over a closed surface S is equal to the Volume integral of the Divergence of vector field $\vec{A}$ taken over the Volume enclosed by the surface.

(b) Let $\vec{A} = A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}$

$$\begin{aligned} \nabla \cdot \vec{A} &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) \\ &= \frac{\partial A_1}{\partial x} + \frac{\partial A_2}{\partial y} + \frac{\partial A_3}{\partial z} \end{aligned}$$

Let $\vec{n} = n_1 \hat{i} + n_2 \hat{j} + n_3 \hat{k}$ be positive unit Normal to S

$$\Rightarrow \text{then } n_1 = \vec{n} \cdot \hat{i} = \cos \alpha ; n_2 = \vec{n} \cdot \hat{j} = \cos \beta ; n_3 = \vec{n} \cdot \hat{k} = \cos \gamma$$

where α, β and γ are the angles which $\vec{n}$ makeswith x -axis, y -axis and z -axis respectively. so $\cos \alpha$, $\cos \beta$ and $\cos \gamma$ are the direction Cosines of $\vec{n}$

$$\begin{aligned} \text{As, } \vec{A} \cdot \vec{n} &= (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) \cdot (\cos \alpha \hat{i} + \cos \beta \hat{j} + \cos \gamma \hat{k}) \\ &= A_1 \cos \alpha + A_2 \cos \beta + A_3 \cos \gamma \end{aligned}$$

Hence, The Divergence Theorem in Rectangular form can be

written as

$$-\iiint_V \left(\frac{\partial A_1}{\partial x} + \frac{\partial A_2}{\partial y} + \frac{\partial A_3}{\partial z} \right) dx dy dz = \iint_S (A_1 \cos \alpha + A_2 \cos \beta + A_3 \cos \gamma) ds$$

15) Demonstrate the divergence Theorem physically.

Sol:-

Let $\vec{V}$ be the velocity of fluid elements
 Then

Volume of fluid crossing ds in time

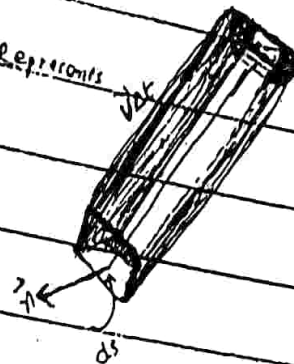
$$\Delta t \text{ is } = \vec{V} \Delta t \cdot \vec{n} ds$$

Volume per second of fluid crossing ds is $= \vec{V} \cdot \vec{n} ds$

Total volume per second of fluid emerging from an closed surface S is $= \iint_S \vec{V} \cdot \vec{n} ds$

Total volume per second of fluid emerging from an volume elements of S is $= \iiint_V \vec{\nabla} \cdot \vec{V} dv$

$$\text{As } \iint_S \vec{V} \cdot \vec{n} ds = \iiint_V \vec{\nabla} \cdot \vec{V} dv$$



16) Prove the divergence Theorem.

Sol:-

Let S be a closed surface

such that any line parallel to coordinate axes cuts S at most two points.

Sub-divide the surface S

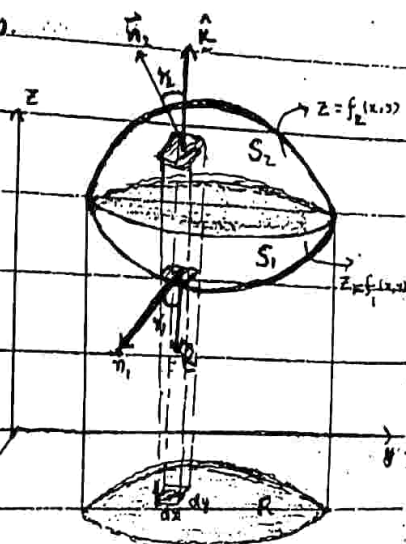
into two portions S_1 and S_2

such that the equations of upper and lower

portions S_1 and S_2 are $z = f_1(x, y)$ and $z = f_2(x, y)$

respectively.

Let R be the projection of surface S on the xy -plane



Now,

$$\begin{aligned}
 \iiint_V \frac{\partial A_3}{\partial z} dV &= \iint_R \left(\int_{f_1(x,y)}^{f_2(x,y)} \frac{\partial A_3}{\partial z} dz \right) dx dy \\
 &= \iint_R \left(A_3(x,y,z) \Big|_{z=f_1(x,y)}^{f_2(x,y)} \right) dx dy \\
 &= \iint_R (A_3(x,y,f_2(x,y)) - A_3(x,y,f_1(x,y))) dx dy \\
 &= \iint_R A_3(x,y,f_2(x,y)) dx dy - \iint_R A_3(x,y,f_1(x,y)) dx dy \quad \rightarrow (1)
 \end{aligned}$$

For the upper portion S_2 :-

$$\begin{aligned}
 ds_2 &= \frac{dxdy}{\cos \gamma_2} = \frac{dxdy}{(\vec{n}_2 \cdot \hat{k})} \\
 \Rightarrow dxdy &= \vec{n}_2 \cdot \hat{k} ds_2 = \hat{k} \cdot \vec{n}_2 ds_2
 \end{aligned}$$

where γ_2 is acute angle which $\vec{n}_2$ makes with $\hat{k}$ For the lower portion S_1 :-

$$\begin{aligned}
 ds_1 &= \frac{dxdy}{-\cos \gamma_1} = \frac{dxdy}{-(\vec{n}_1 \cdot \hat{k})} \\
 \Rightarrow dxdy &= -\vec{n}_1 \cdot \hat{k} ds_1 = -\hat{k} \cdot \vec{n}_1 ds_1
 \end{aligned}$$

where γ_1 is obtuse angle which $\vec{n}_1$ makes with $\hat{k}$

The equation (1) becomes,

$$\begin{aligned}
 \iiint_V \frac{\partial A_3}{\partial z} dV &= \iint_{S_2} A_3(x,y,z) \hat{k} \cdot \vec{n}_2 ds_2 + \iint_{S_1} A_3(x,y,z) \hat{k} \cdot \vec{n}_1 ds_1 \\
 &= \iint_{S_2} A_3(x,y,z) \hat{k} \cdot \vec{n}_2 ds_2 + \iint_{S_1} A_3(x,y,z) \hat{k} \cdot \vec{n}_1 ds_1 \\
 &= \iint_S A_3(x,y,z) \hat{k} \cdot \vec{n} ds \quad \rightarrow (2)
 \end{aligned}$$

Similarly By projecting S on the yz and xz coordinate plane

$$\iiint_V \frac{\partial A_2}{\partial y} dV = \iint_S A_2(x,y,z) \hat{j} \cdot \vec{n} ds \quad \rightarrow (3)$$

$$\iiint_V \frac{\partial A_1}{\partial x} dV = \iint_S A_1(x,y,z) \hat{i} \cdot \vec{n} ds \quad \rightarrow (4)$$

Adding equations (2), (3) and (4)

$$\iiint_V \left(\frac{\partial A_1}{\partial x} + \frac{\partial A_2}{\partial y} + \frac{\partial A_3}{\partial z} \right) dV = \iint_S (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) \cdot \vec{n} \, dS$$

$$\iiint_V (\vec{\nabla} \cdot \vec{A}) \, dV = \iint_S \vec{A} \cdot \vec{n} \, dS = \iiint_S \vec{A} \cdot \vec{ds}$$

Q. Evaluate $\iint_S \vec{F} \cdot \vec{n} \, dS$ where $\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$ and S is the surface of the cube bounded by $x=0, x=1, y=0, y=1, z=0, z=1$.

Sol:-

Given that

$$\vec{F} = 4xz\hat{i} - y^2\hat{j} + yz\hat{k}$$

By Divergence Theorem

$$\iiint_V \vec{\nabla} \cdot \vec{F} \, dV = \iint_S \vec{F} \cdot \vec{n} \, dS$$

$$\iint_S \vec{F} \cdot \vec{n} \, dS = \iiint_V \left(\frac{\partial}{\partial x}(4xz) + \frac{\partial}{\partial y}(-y^2) + \frac{\partial}{\partial z}(yz) \right) dV$$

$$= \iiint_V (4z - 2y + y) \, dV$$

$$= \iiint_V (4z - y) \, dV$$

$$= \int_{x=0}^1 \int_{y=0}^1 \int_{z=0}^1 (4z - y) \, dz \, dy \, dx$$

$$= \int_{x=0}^1 \int_{y=0}^1 \left(2z^2 - yz \right) \Big|_0^1 \, dy \, dx$$

$$= \int_{x=0}^1 \int_{y=0}^1 (2 - y) \, dy \, dx$$

$$= \int_{x=0}^1 \left[2y - \frac{y^2}{2} \right]_0^1 \, dx$$

$$= \int_{x=0}^1 \left(2 - \frac{1}{2} \right) \, dx$$

$$= \int_{x=0}^1 \frac{3}{2} \, dx$$

$$= \frac{3}{2} (1-0)$$

$$= \frac{3}{2}$$

(10) Verify the Divergence Theorem for $\vec{A} = 4xz\hat{i} - 2y^2\hat{j} + z^2\hat{k}$ taken over the Region bounded by $x^2 + y^2 = 4$, $z=0$ and $z=3$

Sol:-

Given that

$$\vec{A} = 4xz\hat{i} - 2y^2\hat{j} + z^2\hat{k}$$

Now

$$\iiint_V \vec{\nabla} \cdot \vec{A} \, dV = \iiint_V \left(\frac{\partial}{\partial x}(4xz) + \frac{\partial}{\partial y}(-2y^2) + \frac{\partial}{\partial z}(z^2) \right) dV$$

$$= \iiint_V (4 - 4y + 2z) \, dV \quad \text{--- (1)}$$

or

$$x^2 + y^2 = 4$$

$$y^2 = 4 - x^2$$

$$y = \pm \sqrt{4 - x^2}$$

For x -intercept put $y=0$

$$x^2 = 4$$

$$x = \pm 2$$

x varies from -2 to 2

y varies from $-\sqrt{4-x^2}$ to $\sqrt{4-x^2}$

z varies from 0 to 3

The equation (1) becomes

$$\iiint_V \vec{\nabla} \cdot \vec{A} \, dV = \int_{x=-2}^2 \int_{y=-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{z=0}^3 (4 - 4y + 2z) \, dz \, dy \, dx$$

$$= \int_{x=-2}^2 \int_{y=-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \left[(4z - 4yz + z^2) \Big|_0^3 \right] dy \, dx$$

$$= \int_{x=-2}^2 \int_{y=-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (12 - 12y + 9) \, dy \, dx$$

$$= \int_{x=-2}^2 \int_{y=-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (21 - 12y) \, dy \, dx$$

$$\begin{aligned}
 \iiint_V \vec{\nabla} \cdot \vec{A} \, dV &= \int_{x=1}^2 \int_{y=1}^2 \int_{z=1}^2 (2xy - 6yz) \, dz \, dy \, dx \\
 &= \int_{x=1}^2 \int_{y=1}^2 \left[2xy(2-1) - 6y(2-1) \right] dy \, dx \\
 &= \int_{x=1}^2 \int_{y=1}^2 (2xy - 6y) \, dy \, dx \\
 &= \int_{x=1}^2 \left[2x \left(\frac{y^2}{2} \right) - 6 \left(\frac{y^2}{2} \right) \right]_{y=1}^2 dx \\
 &= \int_{x=1}^2 \left[2x(2-1) - 3(4-1) \right] dx \\
 &= \int_{x=1}^2 (2x - 9) \, dx \\
 &= \left[x^2 - 9x \right]_{x=1}^2 \\
 &= (4 - 18) - (1 - 9) \\
 &= -14 - (-8) \\
 &= -6
 \end{aligned}$$

Let $x = 2 \sin \theta$

$dx = 2 \cos \theta \, d\theta$

When $x=0$ then $\theta=0$

When $x=2$ then $\theta = \pi/2$

The equation (2) becomes

$$\begin{aligned}
 \iiint_V \vec{\nabla} \cdot \vec{A} \, dV &= 84 \int_{\theta=0}^{\pi/2} \int_{\phi=0}^{\pi/2} \int_{\psi=0}^{\pi/2} 2 \cos \theta \cdot 2 \cos \phi \, d\psi \, d\phi \, d\theta \\
 &= 84 \times 4 \int_{\theta=0}^{\pi/2} \cos^2 \theta \, d\theta \\
 &= 84 \times 4 \int_{\theta=0}^{\pi/2} \frac{1 + \cos 2\theta}{2} \, d\theta \\
 &= 84 \times 2 \left(\theta + \frac{\sin 2\theta}{2} \right) \Big|_0^{\pi/2} \\
 &= 84 \times 2 \left(\frac{\pi}{2} - 0 - 0 + 0 \right) \\
 &= 84 \times \pi \\
 &= 84\pi \quad \rightarrow (A)
 \end{aligned}$$

Now $\vec{A} = 4x^2 \hat{i} - 2y^2 \hat{j} + z^2 \hat{k}$

For $(x^2 + y^2 = 4)$:-

Let $\phi(x, y, z) = x^2 + y^2 - 4$

$\vec{\nabla} \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (x^2 + y^2 - 4)$

$= 2x \hat{i} + 2y \hat{j}$

$|\vec{\nabla} \phi| = \sqrt{4x^2 + 4y^2}$

$$|\nabla\phi| = \sqrt{4(x^2+y^2)}$$

$$= \sqrt{4(4)}$$

$$= 4$$

$$\vec{n} = \frac{\nabla\phi}{|\nabla\phi|}$$

$$= \frac{2x\hat{i} + 2y\hat{j}}{4}$$

$$= \frac{x\hat{i} + y\hat{j}}{2}$$

$$|\vec{n} \cdot \hat{j}| = \frac{y}{2}$$

$$\text{Now } \vec{A} \cdot \vec{n} = (4x\hat{i} - 2y^2\hat{j} + z^2\hat{k}) \cdot \left(\frac{x\hat{i} + y\hat{j}}{2} \right)$$

$$= \frac{4x^2 - 2y^3}{2}$$

$$= 2x^2 - y^3$$

As,

$$\iint_{S_1} \vec{A} \cdot \vec{n} \, dS_1 = \iint_R (2x^2 - y^3) \cdot \frac{dx \, dz}{|\vec{n} \cdot \hat{j}|}$$

$$= \iint_R (2x^2 - y^3) \cdot \frac{dx \, dz}{y/2}$$

$$= 2 \iint_R \left(\frac{2x^2}{y} - y^2 \right) dx \, dz$$

Alternatively,

$$x^2 + y^2 = 4$$

Replace

$x = 2\cos\theta$ &

$y = 2\sin\theta$ so,

$dx \, dy = 2 \, d\theta \, dz$

then $0 \leq \theta \leq 2\pi$, $0 \leq z \leq 3$

$\oint \vec{A} \cdot \vec{n} \, dS_1$

$$= \int_0^{2\pi} \int_0^3 \left(\frac{2(2\cos\theta)^2}{2\sin\theta} - (2\sin\theta)^2 \right) 2 \, d\theta \, dz$$

$$= \int_0^{2\pi} \left(4\cos^2\theta - 4\sin^2\theta \right) 2 \, d\theta$$

$$= \int_0^{2\pi} 4\cos(2\theta) \, d\theta$$

$$= 4\sin(2\theta) \Big|_0^{2\pi} = 0$$

$$y^2 = 4 - x^2 \Rightarrow y = \pm\sqrt{4-x^2}$$

$$\iint_{S_1} \vec{A} \cdot \vec{n} \, dS_1 = 2 \times 2 \iint_R \frac{2x^2}{\sqrt{4-x^2}} - (4-x^2) \, dx \, dz \quad (\text{From symmetry multiply by 2})$$

$$= 4 \iint_R \frac{-2(4-x^2-4) - (4-x^2)}{\sqrt{4-x^2}} \, dx \, dz$$

$$= 4 \iint_R \left[-2\sqrt{4-x^2} + 8(4-x^2)^{-1/2} - (4-x^2) \right] dx \, dz \rightarrow (3)$$

$$x^2 + y^2 = 4$$

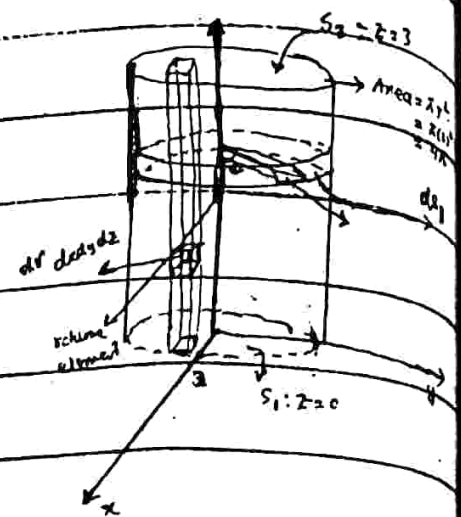
For x -intercept, $y=0$

$$x^2 = 4$$

$$x = \pm 2$$

As x varies from -2 to 2

z varies from 0 to 3



The equation (1) becomes

$$\begin{aligned} \iint_{S_1} \vec{A} \cdot \vec{n} \, ds_1 &= 2 \times 9 \int_{x=-2}^2 \int_{z=0}^3 \left[-2\sqrt{4-x^2} + \frac{8}{\sqrt{4-x^2}} - (4-x^2) \right] dz \, dx \\ &= 4 \int_{x=-2}^2 \left[-6\sqrt{4-x^2} + \frac{24}{\sqrt{4-x^2}} - 3(4-x^2) \right] dx \\ &= 4 \int_{x=-2}^2 \left[-6\sqrt{4-x^2} + \frac{24}{\sqrt{4-x^2}} - 3(4-x^2) \right] dx \end{aligned}$$

$$\begin{aligned} &= -24 \int_{x=-2}^2 \sqrt{4-x^2} \, dx + 96 \int_{x=-2}^2 \frac{1}{\sqrt{4-x^2}} \, dx - 3 \left(4x - \frac{x^3}{3} \right) \Big|_{-2}^2 \\ &= -24 \int_{x=-2}^2 \sqrt{4-x^2} \, dx + 96 \int_{x=-2}^2 \frac{1}{\sqrt{4-x^2}} \, dx - 3a \quad \text{--- (2)} \end{aligned}$$

Let $x = 2 \sin \theta \Rightarrow dx = 2 \cos \theta \, d\theta$

As, $-2 \leq x \leq 2$ Then $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

The equation (2) becomes

$$\begin{aligned} \iint_{S_1} \vec{A} \cdot \vec{n} \, ds_1 &= -24 \int_{-\pi/2}^{\pi/2} 4 \cos^2 \theta \, d\theta + 96 \int_{-\pi/2}^{\pi/2} d\theta - 32 \int_{-\pi/2}^{\pi/2} 1 \, d\theta \\ &= -96 \int_{-\pi/2}^{\pi/2} \cos^2 \theta \, d\theta + 96 \left(\frac{\pi}{2} - \frac{-\pi}{2} \right) - 32 \left(\frac{\pi}{2} - \frac{-\pi}{2} \right) \\ &= 48 \quad (\because -32 \text{ is cancelled out}) \end{aligned}$$

For $z=0$: As $\vec{n} = -\hat{k}$ and $z=0$

So, $\vec{A} \cdot \vec{n} = (4x^2\hat{i} - 2y^2\hat{j} + 0\hat{k}) \cdot -\hat{k} = 0$

As, $\iint_{S_2} \vec{A} \cdot \vec{n} \, ds_2 = 0$

For $z=3$:

As, $\vec{n} = \hat{k}$ and $z=3$

So, $\vec{A} \cdot \vec{n} = (4x^2\hat{i} - 2y^2\hat{j} + 9\hat{k}) \cdot \hat{k} = 9$

As, $\iint_{S_3} \vec{A} \cdot \vec{n} \, ds_3 = \iint_{S_3} 9 \, ds_3 = 9 \iint_{S_3} ds_3 = 9(4\pi) = 36\pi$

So,

$$\iint_S \vec{A} \cdot \vec{n} \, ds = \iint_{S_1} \vec{A} \cdot \vec{n} \, ds_1 + \iint_{S_2} \vec{A} \cdot \vec{n} \, ds_2 + \iint_{S_3} \vec{A} \cdot \vec{n} \, ds_3$$

$$= 48\pi + 36\pi + 0$$

$$= 84\pi \quad \text{--- (B)}$$

Therefore, From equation (A) and (B)

$$\iiint_V \vec{\nabla} \cdot \vec{A} \, dV = \iint_S \vec{A} \cdot \vec{n} \, ds$$

Alternatively,

$$\iint_{S_1} ds_1 = \int_0^{2\pi} \int_0^{\pi/2} r \, dr \, d\theta$$

$$\text{Let } x = r \cos \theta, y = r \sin \theta$$

$$dx = -r \sin \theta \, d\theta, dy = r \cos \theta \, d\theta$$

$$\therefore \frac{1}{2} \int_0^{2\pi} (4r^2 \sin^2 \theta + 4r^2 \cos^2 \theta) \, d\theta$$

$$= \frac{1}{2} \int_0^{2\pi} 4r^2 \, d\theta$$

$$= 2 \int_0^{2\pi} r^2 \, d\theta$$

$$= 4\pi$$

$$\text{or } \iint_{S_3} ds_3 = \pi r^2$$

$$= \pi \cdot 2^2 = 4\pi$$

20) Evaluate $\iint_S \vec{r} \cdot \vec{n} \, ds$ where S is a closed surface.

Sol:-

let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

Then By divergence theorem

$$\begin{aligned} \iint_S \vec{r} \cdot \vec{n} \, ds &= \iiint_V \vec{\nabla} \cdot \vec{r} \, dv \\ &= \iiint_V \left(\frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) \right) dv \\ &= \iiint_V (1+1+1) \, dv \\ &= 3 \iiint_V dv \\ &= 3V \end{aligned}$$

GREEN'S First And Second Identity (or Symmetrical Theorem)

21) Prove $\iiint_V (\phi \nabla^2 \psi - \psi \nabla^2 \phi) \, dv = \iint_S (\phi \nabla \psi - \psi \nabla \phi) \cdot \vec{ds}$

Sol:-

let $\vec{A} = \phi \nabla \psi$

By Divergence Theorem

$$\iiint_V \vec{\nabla} \cdot \vec{A} \, dv = \iint_S \vec{A} \cdot \vec{n} \, ds = \iint_S \vec{A} \cdot \vec{ds}$$

As,

$$\iiint_V \vec{\nabla} \cdot (\phi \nabla \psi) \, dv = \iint_S \phi \nabla \psi \cdot \vec{ds}$$

$$\iiint_V [\phi \nabla^2 \psi + (\nabla \phi) \cdot (\nabla \psi)] \, dv = \iint_S \phi \nabla \psi \cdot \vec{ds} \quad \text{--- (Green First Identity)}$$

Interchanging ϕ by ψ

$$\iiint_V [\psi \nabla^2 \phi + (\nabla \psi) \cdot (\nabla \phi)] \, ds = \iint_S \psi \nabla \phi \cdot \vec{ds} \quad \text{--- (2)}$$

Subtracting (2) from (1)

$$\begin{aligned} \iiint_V [\phi \nabla^2 \psi + (\nabla \phi) \cdot (\nabla \psi) - \psi \nabla^2 \phi - (\nabla \psi) \cdot (\nabla \phi)] \, dv &= \iint_S (\phi \nabla \psi - \psi \nabla \phi) \cdot \vec{ds} \\ \iiint_V [\phi \nabla^2 \psi - \psi \nabla^2 \phi] \, dv &= \iint_S (\phi \nabla \psi - \psi \nabla \phi) \cdot \vec{ds} \end{aligned}$$

which is The Green's Second Identity.

Prove $\iiint_V \nabla \phi \, dv = \iint_S \phi \vec{n} \, ds$ (Gradient Theorem)

(22)

Sol:-

Let $\vec{A} = \phi \vec{c}$ where $\vec{c}$ is a constant vector

By Divergence Theorem

$$\iiint_V \nabla \cdot \vec{A} \, dv = \iint_S \vec{A} \cdot \vec{n} \, ds$$

$$\iiint_V \nabla \cdot (\phi \vec{c}) \, dv = \iint_S \phi \vec{c} \cdot \vec{n} \, ds$$

As,

$$\iiint_V \nabla \phi \cdot \vec{c} \, dv = \iint_S \vec{c} \cdot (\phi \vec{n}) \, ds$$

$$\iiint_V \vec{c} \cdot \nabla \phi \, dv = \iint_S \vec{c} \cdot (\phi \vec{n}) \, ds$$

$$\vec{c} \cdot \iiint_V \nabla \phi \, dv = \vec{c} \cdot \iint_S \phi \vec{n} \, ds$$

$$\iiint_V \nabla \phi \, dv = \iint_S \phi \vec{n} \, ds$$

(23)

Prove $\iiint_V \nabla \times \vec{B} \, dv = \iint_S \vec{n} \times \vec{B} \, ds$

(CURL THEOREM)

Sol:-

Let $\vec{A} = \vec{B} \times \vec{c}$ where $\vec{c}$ is a constant vector

By Divergence Theorem

$$\iiint_V \nabla \cdot \vec{A} \, dv = \iint_S \vec{A} \cdot \vec{n} \, ds$$

$$\iiint_V \nabla \cdot (\vec{B} \times \vec{c}) \, dv = \iint_S (\vec{B} \times \vec{c}) \cdot \vec{n} \, ds$$

$$\iiint_V \vec{c} \cdot (\nabla \times \vec{B}) \, dv = \iint_S \vec{c} \cdot (\vec{n} \times \vec{B}) \, ds$$

$$\vec{c} \cdot \iiint_V (\nabla \times \vec{B}) \, dv = \vec{c} \cdot \iint_S \vec{n} \times \vec{B} \, ds$$

so,

$$\iiint_V (\nabla \times \vec{B}) \, dv = \iint_S \vec{n} \times \vec{B} \, ds$$

Alternatively,

$$\iiint_V \nabla \cdot \vec{A} \, dv = \iint_S \vec{A} \cdot \vec{n} \, ds$$

$$\iiint_V \nabla \cdot (\vec{B} \times \vec{c}) \, dv = \iint_S (\vec{B} \times \vec{c}) \cdot \vec{n} \, ds$$

$$\iiint_V \vec{c} \cdot (\nabla \times \vec{B}) \, dv = \iint_S \vec{c} \cdot (\vec{n} \times \vec{B}) \, ds$$

$$\iiint_V \nabla \times \vec{B} \, dv = \iint_S \vec{n} \times \vec{B} \, ds$$

$$\iiint_V \nabla \times \vec{B} \, dv = \iint_S \vec{n} \times \vec{B} \, ds$$

$$\iiint_V \nabla \times \vec{B} \, dv = \iint_S \vec{n} \times \vec{B} \, ds$$

GAUSS' THEOREM:-

- (26) Let S be a closed surface and let $\vec{r}$ denote the position vector of any point (x, y, z) measured from an origin O . Prove that

$$\iint_S \frac{\vec{n} \cdot \vec{r}}{r^3} ds$$

is equal to

(a) Zero if origin O lies outside S

(b) 4π if origin O lies inside S

Sol:-

(a) By Divergence Theorem

$$\iint_S \frac{\vec{n} \cdot \vec{r}}{r^3} ds = \iiint_V \vec{\nabla} \cdot \frac{\vec{r}}{r^3} dv$$

we know

$$\vec{\nabla} \cdot \frac{\vec{r}}{r^3} = 0 \quad \text{if } r \neq 0$$

So, $\vec{\nabla} \cdot \frac{\vec{r}}{r^3} = 0$ if origin O lies outside S

Therefore

$$\iint_S \frac{\vec{n} \cdot \vec{r}}{r^3} ds = 0 \quad \text{if origin } O \text{ lies outside } S$$

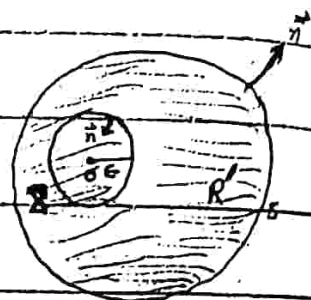
(b) If origin O lies inside S , surround

O by a small sphere Σ with

center at O and radius ϵ

Let R' be the region bounded by

S and Σ



then by divergence Theorem

$$\iint_{S+\Sigma} \frac{\vec{n} \cdot \vec{r}}{r^3} ds = \iiint_{R'} \vec{\nabla} \cdot \frac{\vec{r}}{r^3} dv = 0 \quad (\because r \neq 0 \text{ in } R')$$

$$\iint_S \frac{\vec{n} \cdot \vec{r}}{r^3} ds + \iint_{\Sigma} \frac{\vec{n} \cdot \vec{r}}{r^3} ds = 0$$

$$\iint_S \frac{\vec{n} \cdot \vec{r}}{r^3} ds = - \iint_{\Sigma} \frac{\vec{n} \cdot \vec{r}}{r^3} ds \rightarrow \textcircled{1}$$

Now on sphere Σ

$$r = e \text{ and } \vec{n} = -\frac{\vec{r}}{r} = -\frac{\vec{r}}{e} = -\frac{\vec{r}}{e}$$

$$\frac{\vec{n} \cdot \vec{r}}{r^3} = \frac{-\vec{r} \cdot \vec{r}}{e \cdot e^3} = \frac{-r^2}{e^4} = \frac{-e^2}{e^4} = \frac{-1}{e^2}$$

on Σ

$$r = e ; -\vec{r} = \vec{n}$$

then

$$\frac{\vec{n} \cdot \vec{r}}{r^3} = \frac{-\vec{r} \cdot \vec{r}}{r^3} = \frac{-r^2}{r^3} = \frac{-1}{r} = \frac{-1}{e}$$

The Equation (1) becomes

$$\iint_{\Sigma} \frac{\vec{n} \cdot \vec{r}}{r^3} ds = - \iint_{\Sigma} \frac{1}{e^2} ds$$

$$= - \frac{1}{e^2} \iint_{\Sigma} ds$$

$$= - \frac{1}{e^2} \cdot 4\pi e^2$$

$$= -4\pi \text{ when origin } O \text{ lies inside } \Sigma$$

$$\pi - \pi - \pi$$

STOKES' THEOREM..

(30) (a) Express Stokes' Theorem in words

and

(b) Write its Rectangular form.

Soln.

(a) As,

$$\oint_C \vec{A} \cdot d\vec{r} = \iint_S (\vec{\nabla} \times \vec{A}) \cdot \vec{n} \, ds = \iint_S (\vec{\nabla} \times \vec{A}) \cdot \vec{ds}$$

The Line Integral of the Tangential component of a vector field $\vec{A}$ taken around a simple closed curve C is equal to the Surface integral of the Normal component of Curl of the vector field $\vec{A}$ taken over any surface S having C as its boundary.

(b) Let $\vec{A} = A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}$

$$d\vec{r} = dx \hat{i} + dy \hat{j} + dz \hat{k}$$

$$\text{As } \vec{A} \cdot d\vec{r} = (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k}) \cdot (dx \hat{i} + dy \hat{j} + dz \hat{k})$$

$$= A_1 dx + A_2 dy + A_3 dz$$

$$\text{Now } \vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_1 & A_2 & A_3 \end{vmatrix}$$

$$= \left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) \hat{i} + \left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) \hat{j} + \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \hat{k}$$

Let $\vec{n} = n_1 \hat{i} + n_2 \hat{j} + n_3 \hat{k}$ be unit Normal to S

$$\Rightarrow n_1 = \vec{n} \cdot \hat{i} = \cos \alpha, \quad n_2 = \vec{n} \cdot \hat{j} = \cos \beta, \quad n_3 = \vec{n} \cdot \hat{k} = \cos \gamma$$

where α, β and γ are the angles which $\vec{n}$ makes with x -axis, y -axis and z -axis respectively. As $\cos \alpha$, $\cos \beta$ and $\cos \gamma$ are the direction cosines of $\vec{n}$.

$$(\vec{\nabla} \times \vec{A}) \cdot \vec{n} = \left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) \cos \alpha + \left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) \cos \beta + \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \cos \gamma$$

Hence the Stokes' Theorem in Rectangular form can be written as,

$$\oint_C A_1 dx + A_2 dy + A_3 dz = \iint_S \left[\left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z} \right) \cos \alpha + \left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x} \right) \cos \beta + \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y} \right) \cos \gamma \right] ds$$

* — * — *

Q1) Prove Stokes Theorem.

Sol:-

Let "S" be a surface such that its projections on the xy , yz and zx planes are regions bounded by simple closed curves.

Assume S to have representation

$$z = f(x, y) \text{ or } y = g(x, z) \text{ or } x = h(y, z)$$

where f, g, h are single

valued, continuous and differentiable functions.

Consider

$$\iint_S (\vec{\nabla} \times (A_1 \hat{i})) \cdot \vec{n} ds$$

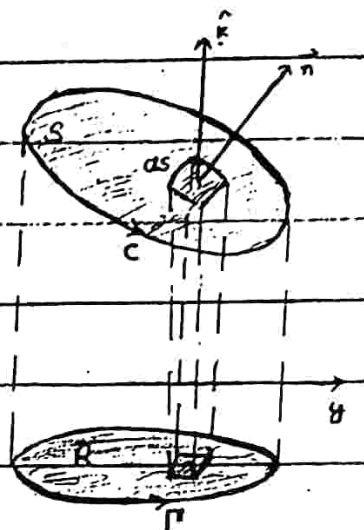
Since $\vec{\nabla} \times (A_1 \hat{i}) =$

$\hat{i}$	$\hat{j}$	$\hat{k}$
$\frac{\partial}{\partial x}$	$\frac{\partial}{\partial y}$	$\frac{\partial}{\partial z}$
A_1	0	0

$$= \frac{\partial A_1}{\partial z} \hat{j} - \frac{\partial A_1}{\partial y} \hat{k}$$

$$[\vec{\nabla} \times (A_1 \hat{i})] \cdot \vec{n} = \left(\frac{\partial A_1}{\partial z} \hat{j} - \frac{\partial A_1}{\partial y} \hat{k} \right) \cdot \vec{n}$$

$$[\vec{\nabla} \times (A_1 \hat{i})] \cdot \vec{n} ds = \left[\frac{\partial A_1}{\partial z} \vec{n} \cdot \hat{j} - \frac{\partial A_1}{\partial y} \vec{n} \cdot \hat{k} \right] ds \rightarrow (1)$$



Let $z = f(x, y)$ be the equation of S then the

position vector $\vec{r}$ at any point of S is

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\vec{r} = x\hat{i} + y\hat{j} + f(x, y)\hat{k}$$

$$\frac{\partial \vec{r}}{\partial y} = \hat{j} + \frac{\partial f}{\partial y} \hat{k} = \hat{j} + \frac{\partial z}{\partial y} \hat{k}$$

But $\frac{\partial \vec{r}}{\partial y}$ is a vector tangent to S and thus perpendicular to $\vec{n}$ so,

$$\vec{n} \cdot \frac{\partial \vec{r}}{\partial y} = 0$$

$$\vec{n} \cdot \left(\hat{j} + \frac{\partial z}{\partial y} \hat{k} \right) = 0$$

$$\vec{n} \cdot \hat{j} + \frac{\partial z}{\partial y} \vec{n} \cdot \hat{k} = 0$$

$$\vec{n} \cdot \hat{j} = - \frac{\partial z}{\partial y} \vec{n} \cdot \hat{k}$$

The Equation (1) becomes,

$$[\vec{\nabla} \times (A_1 \hat{i})] \cdot \vec{n} dS = \left[- \frac{\partial A_1}{\partial z} \frac{\partial z}{\partial y} \vec{n} \cdot \hat{k} - \frac{\partial A_1}{\partial y} \vec{n} \cdot \hat{k} \right] dS$$

$$= - \left(\frac{\partial A_1}{\partial z} \frac{\partial z}{\partial y} + \frac{\partial A_1}{\partial y} \right) \vec{n} \cdot \hat{k} dS$$

$$= - \left(\frac{\partial A_1}{\partial z} + \frac{\partial A_1}{\partial y} \frac{\partial z}{\partial y} \right) \vec{n} \cdot \hat{k} dS$$

Now on S $A_1(x, y, z) = A_1(x, y, f(x, y)) = F(x, y)$

$$\text{As, } \frac{\partial A_1}{\partial y} + \frac{\partial A_1}{\partial z} \frac{\partial z}{\partial y} = \frac{\partial F}{\partial y}$$

So,

$$[\vec{\nabla} \times (A_1 \hat{i})] \cdot \vec{n} dS = - \frac{\partial F}{\partial y} dx dy$$

$$\therefore dS = \frac{dx dy}{\hat{n} \cdot \hat{i}}$$

As,

$$\iint_S [\vec{\nabla} \times (A_1 \hat{i})] \cdot \vec{n} dS = \iint_R - \frac{\partial F}{\partial y} dx dy \rightarrow (2) \text{ where } R \text{ is projection of } S \text{ on } xy\text{-plane}$$

By Green's Theorem in the plane

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

For $N = 0$

$$\oint M dx = \iint_R - \frac{\partial M}{\partial y} dx dy$$

As, $\oint_{\Gamma} F dx = \iint_R -\frac{\partial F}{\partial y} dx dy$

where Γ is the boundary of R .

And $\oint_{\Gamma} F dx = \oint_C A_1 dx$

The equation (2) becomes

$$\iint_S [\nabla \times (A_1 \hat{i})] \cdot \vec{n} ds = \oint_C A_1 dx \rightarrow (3)$$

Similarly, By projecting S on yz and zx coordinate planes

$$\iint_S [\nabla \times (A_1 \hat{j})] \cdot \vec{n} ds = \oint_C A_2 dy \rightarrow (4)$$

$$\iint_S [\nabla \times (A_1 \hat{k})] \cdot \vec{n} ds = \oint_C A_3 dz \rightarrow (5)$$

Adding Equations (3), (4) and (5)

$$\iint_S [\nabla \times (A_1 \hat{i})] \cdot \vec{n} ds + \iint_S [\nabla \times (A_2 \hat{j})] \cdot \vec{n} ds + \iint_S [\nabla \times (A_3 \hat{k})] \cdot \vec{n} ds$$

$$= \oint_C A_1 dx + \oint_C A_2 dy + \oint_C A_3 dz$$

$$\iint_S [\nabla \times (A_1 \hat{i} + A_2 \hat{j} + A_3 \hat{k})] \cdot \vec{n} ds = \oint_C A_1 dx + A_2 dy + A_3 dz$$

$$\iint_S (\nabla \times \vec{A}) \cdot \vec{n} ds = \oint_C \vec{A} \cdot d\vec{r}$$

Hence proved

* * *

(32) verify Stokes' Theorem for $\vec{A} = (2x - y)\hat{i} - yz^2\hat{j} - y^2z\hat{k}$, where

S is the upper half surface of the sphere $x^2 + y^2 + z^2 = 1$

and C is its boundary.

Sol:-

Given that

$$\vec{A} = (2x - y)\hat{i} - yz^2\hat{j} - y^2z\hat{k}$$

and $x^2 + y^2 + z^2 = 1$

For the upper half surface $z \geq 0$

so, $x^2 + y^2 = 1$

Now

$$\oint_C \vec{A} \cdot d\vec{r} = \oint_C [(2x-y)\hat{i} - y\hat{j} - yz\hat{k}] \cdot [dx\hat{i} + dy\hat{j} + dz\hat{k}]$$

$$= \oint_C (2x-y)dx \quad \dots \textcircled{1}$$

As,

$$x^2 + y^2 = 1$$

where

$$x = \cos \theta$$

$$y = \sin \theta$$

$$0 \leq \theta < 2\pi$$

$$dx = -\sin \theta d\theta$$

$$dy = \cos \theta d\theta$$

This equation $\textcircled{1}$ becomes

$$\oint_C \vec{A} \cdot d\vec{r} = \int_0^{2\pi} (2\cos \theta - \sin \theta) (-\sin \theta d\theta)$$

$$= \int_0^{2\pi} (-2\sin \theta + \sin^2 \theta) d\theta$$

$$= -2 \int_0^{2\pi} \sin \theta d\theta + \int_0^{2\pi} \frac{1 - \cos 2\theta}{2} d\theta$$

$$= -2 \left[-\cos \theta \right]_0^{2\pi} + \frac{1}{2} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{2\pi}$$

$$= 0 + \frac{1}{2} (2\pi - 0 - 0 + 0)$$

$$= \frac{1}{2} (2\pi)$$

$$= \pi$$

 $\textcircled{A}$

As,

$$\vec{A} = (2x-y)\hat{i} - yz^2\hat{j} - y^2z\hat{k}$$

$$\vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x-y & -yz^2 & -y^2z \end{vmatrix}$$

$$= \hat{i} (-2yz + yz) + \hat{j} (0-0) + \hat{k} (0+1)$$

$$= 0 + 0 + \hat{k}$$

$$= \hat{k}$$

Now

$$\iint_S (\vec{\nabla} \times \vec{A}) \cdot \vec{n} dS = \iint_S \hat{k} \cdot \vec{n} dS$$

$$= \iint_S dx dy \rightarrow \textcircled{A}$$

$$\therefore dS = \frac{dx dy}{\hat{k} \cdot \vec{n}}$$

$$x^2 + y^2 = 1$$

For x-Intercept put $y=0$

$$x^2 = 1$$

$$x = \pm 1$$

As $y^2 = 1 - x^2$

$$y = \pm \sqrt{1-x^2}$$

The Equation (2) Becomes

$$\begin{aligned} \iint_S (\nabla \times \vec{A}) \cdot \vec{n} \, ds &= \int_{x=-1}^1 \int_{y=-\sqrt{1-x^2}}^{\sqrt{1-x^2}} dy \, dx \\ &= \int_{x=-1}^1 \left[\sqrt{1-x^2} + \sqrt{1-x^2} \right] dx \\ &= \int_{x=-1}^1 2\sqrt{1-x^2} \, dx \quad \text{--- (3)} \end{aligned}$$

Let $x = \sin \theta$

$$dx = \cos \theta \, d\theta$$

where θ varies from $-\pi/2$ to $\pi/2$

The Equation (3) Becomes

$$\begin{aligned} \iint_S (\nabla \times \vec{A}) \cdot \vec{n} \, ds &= \int_{-\pi/2}^{\pi/2} 2 \cos \theta \cdot \cos \theta \, d\theta \\ &= 2 \int_{-\pi/2}^{\pi/2} \cos^2 \theta \, d\theta \\ &= 2 \int_{-\pi/2}^{\pi/2} (1 + \cos 2\theta) \, d\theta \\ &= \left[2\theta + \sin 2\theta \right]_{-\pi/2}^{\pi/2} \\ &= \frac{\pi}{2} + 0 + \frac{\pi}{2} - 0 \\ &= \pi \quad \text{--- (4)} \end{aligned}$$

Therefore from equation (A) and (B):

$$\oint_C \vec{A} \cdot d\vec{r} = \iint_S (\nabla \times \vec{A}) \cdot \vec{n} \, ds$$

(33) Prove that a necessary and sufficient condition that

$$\oint_C \vec{A} \cdot d\vec{r} = 0 \text{ for every closed curve } C \text{ is that}$$

$$\nabla \times \vec{A} = \vec{0} \text{ identically}$$

Sol:-

Suppose $\oint_C \vec{A} \cdot d\vec{r} = 0$ for every closed curve C

Now we are to show that $\nabla \times \vec{A} = \vec{0}$

We contrary suppose $\nabla \times \vec{A} \neq \vec{0}$ at point P

Then by continuity of $\nabla \times \vec{A} \rightarrow \nabla \times \vec{A} \neq \vec{0}$ in some

Region surrounding P . Let S be a surface lying in this

Region whose Normal $\vec{n}$ at each point is parallel to $\nabla \times \vec{A}$

Then for any scalar α , $\nabla \times \vec{A} = \alpha \vec{n}$ of C is the boundary

S . Then by Stokes' Theorem

$$\oint_C \vec{A} \cdot d\vec{r} = \iint_S (\nabla \times \vec{A}) \cdot \vec{n} \, ds = \iint_S \alpha \vec{n} \cdot \vec{n} \, ds > 0$$

This is the contradiction to our supposition is wrong.

$$\text{Hence } \nabla \times \vec{A} = \vec{0}$$

Conversely, suppose $\nabla \times \vec{A} = \vec{0}$

Then by Stokes' Theorem

$$\oint_C \vec{A} \cdot d\vec{r} = \iint_S (\nabla \times \vec{A}) \cdot \vec{n} \, ds$$

$$= 0$$

(34) Prove $\oint_C d\vec{r} \times \vec{B} = \iint_S (\vec{n} \times \nabla) \times \vec{B} \, ds$

Sol:-

Let $\vec{A} = \vec{B} \times \vec{C}$ where $\vec{C}$ is a constant vector

By Stokes' Theorem

$$\oint_C \vec{A} \cdot d\vec{r} = \iint_S (\vec{\nabla} \times \vec{A}) \cdot \vec{n} \, ds$$

$$\oint_C (\vec{B} \times \vec{c}) \cdot d\vec{r} = \iint_S [\vec{\nabla} \times (\vec{B} \times \vec{c})] \cdot \vec{n} \, ds$$

$$\oint_C \vec{c} \cdot d\vec{r} \times \vec{B} = \iint_S [(\vec{\nabla} \cdot \vec{c}) \vec{B} - (\vec{\nabla} \cdot \vec{B}) \vec{c}] \cdot \vec{n} \, ds$$

$$\begin{aligned} \vec{c} \cdot \oint_C d\vec{r} \times \vec{B} &= \iint_S [(\vec{c} \cdot \vec{\nabla}) \vec{B} - \vec{c} (\vec{\nabla} \cdot \vec{B})] \cdot \vec{n} \, ds \\ &= \iint_S (\vec{c} \cdot \vec{\nabla}) \vec{B} \cdot \vec{n} \, ds - \iint_S \vec{c} (\vec{\nabla} \cdot \vec{B}) \cdot \vec{n} \, ds \end{aligned}$$

$$= \iint_S \vec{c} \cdot \vec{\nabla} (\vec{B} \cdot \vec{n}) \, ds - \iint_S \vec{c} \cdot (\vec{\nabla} \cdot \vec{B}) \vec{n} \, ds$$

$$= \vec{c} \cdot \iint_S \vec{\nabla} (\vec{B} \cdot \vec{n}) \, ds - \vec{c} \cdot \iint_S (\vec{\nabla} \cdot \vec{B}) \vec{n} \, ds$$

$$\vec{c} \cdot \oint_C d\vec{r} \times \vec{B} = \vec{c} \cdot \iint_S [\vec{\nabla} (\vec{B} \cdot \vec{n}) - (\vec{\nabla} \cdot \vec{B}) \vec{n}] \, ds$$

$$\oint_C d\vec{r} \times \vec{B} = \iint_S [(\vec{n} \cdot \vec{B}) \vec{\nabla} - (\vec{\nabla} \cdot \vec{B}) \vec{n}] \, ds$$

$$\oint_C d\vec{r} \times \vec{B} = \iint_S (\vec{n} \times \vec{\nabla}) \times \vec{B} \, ds$$

Alternatively,

$$\begin{aligned} \oint_C \vec{A} \cdot d\vec{r} &= \oint_C (\vec{B} \times \vec{c}) \cdot d\vec{r} = \oint_C d\vec{r} \cdot (\vec{B} \times \vec{c}) \\ &= \oint_C (d\vec{r} \times \vec{B}) \cdot \vec{c} = \oint_C \vec{c} \cdot (d\vec{r} \times \vec{B}) \end{aligned}$$

Now

$$\begin{aligned} \iint_S (\vec{\nabla} \times \vec{A}) \cdot \vec{n} \, ds &= \iint_S \vec{n} \cdot (\vec{\nabla} \times \vec{A}) \, ds \\ &= \iint_S (\vec{n} \times \vec{\nabla}) \cdot \vec{A} \, ds \\ &= \iint_S (\vec{n} \times \vec{\nabla}) \cdot (\vec{B} \times \vec{c}) \, ds \\ &= \iint_S ((\vec{n} \times \vec{\nabla}) \times \vec{B}) \cdot \vec{c} \, ds \\ &= \iint_S \vec{c} \cdot ((\vec{n} \times \vec{\nabla}) \times \vec{B}) \, ds \end{aligned}$$

By Stokes' Theorem

$$\begin{aligned} \oint_C \vec{A} \cdot d\vec{r} &= \iint_S (\vec{\nabla} \times \vec{A}) \cdot \vec{n} \, ds \\ \oint_C \vec{c} \cdot (d\vec{r} \times \vec{B}) &= \iint_S \vec{c} \cdot ((\vec{n} \times \vec{\nabla}) \times \vec{B}) \, ds \\ \vec{c} \cdot \oint_C d\vec{r} \times \vec{B} &= \vec{c} \cdot \iint_S ((\vec{n} \times \vec{\nabla}) \times \vec{B}) \, ds \\ \oint_C d\vec{r} \times \vec{B} &= \iint_S ((\vec{n} \times \vec{\nabla}) \times \vec{B}) \, ds \end{aligned}$$

SUPPLEMENTARY PROBLEMS

2. Verify Green's Theorem in the plane for $\oint_C (3x^2 - 8xy^2) dx + (4y - 6xy) dy$ where C is the boundary of the Region defined by

① $y = \sqrt{x}$, $x = x^2$

② $x = 0$, $y = 0$, $x + y = 1$

Sol:-

① Given that

$$\oint_C M dx + N dy = \oint_C (3x^2 - 8xy^2) dx + (4y - 6xy) dy \rightarrow ①$$

As, $y = \sqrt{x}$; $y = x^2$

$$\sqrt{x} = x^2$$

$$x = x^4$$

$$x^4 - x = 0$$

$$x(x^3 - 1) = 0$$

$$x = 0 \quad ; \quad x^3 - 1 = 0$$

$$(x-1)(x^2+x+1) = 0$$

$$x-1 = 0$$

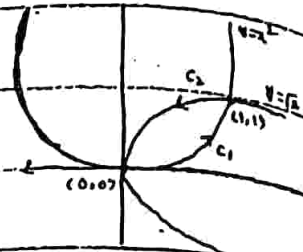
$$x^2+x+1 \neq 0$$

(Not possible)

$$x = 1$$

when $x = 0$ then $y = 0$

when $x = 1$ then $y = 1$



Along $y = \sqrt{x}$

$$\int_C M dx + N dy = - \int_{x=0}^1 (3x^2 - 8x) dx + (4\sqrt{x} - 6x\sqrt{x}) \left(\frac{1}{2\sqrt{x}} dx \right)$$

$$= - \int_{x=0}^1 (3x^2 - 8x + 2 - 3x) dx$$

$$= - \int_{x=0}^1 (3x^2 - 11x + 2) dx$$

$$= - \left(x^3 - \frac{11x^2}{2} + 2x \right) \Big|_0^1$$

$$= - \left(1 - \frac{11}{2} + 2 \right)$$

$$= - 5/2$$

Along $y = x^2$:-

$$\begin{aligned}
 \int_{C_1} M dx + N dy &= \int_{x=0}^1 (3x^2 - 0y^4) dx + (4x^2 - 6x \cdot x^2) (2x dx) \\
 &= \int_{x=0}^1 (3x^2 - 8x^4 + 8x^3 - 12x^4) dx \\
 &= \int_{x=0}^1 (3x^2 - 20x^4 + 8x^3) dx \\
 &= (x^3 - 4x^5 + 2x^4) \Big|_0^1 \\
 &= 1 - 4 + 2 \\
 &= -1
 \end{aligned}$$

The equation (2) becomes

$$\begin{aligned}
 \oint_C M dx + N dy &= \int_{C_1} M dx + N dy + \int_{C_2} M dx + N dy \\
 &= \frac{5}{2} - 1 \\
 &= \frac{3}{2} \rightarrow (2)
 \end{aligned}$$

Since

$$M(x, y) = 3x^2 - 8y^2 \quad ; \quad N(x, y) = 4y - 6xy$$

$$\frac{\partial M}{\partial y} = -16y \quad ; \quad \frac{\partial N}{\partial x} = -6y$$

As,

$$\begin{aligned}
 \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy &= \iint_{x=0, y=x^2}^{x=1, y=x^2} (-6y + 16y) dy dx \\
 &= \int_{x=0}^1 \int_{y=x^2}^{y=x^2} (10y) dy dx \\
 &= \int_{x=0}^1 \left(5y^2 \Big|_{x^2}^{x^2} \right) dx \\
 &= \int_{x=0}^1 5(x - x^4) dx \\
 &= 5 \left(\frac{x^2}{2} - \frac{x^5}{5} \right) \Big|_0^1 \\
 &= 5 \left(\frac{1}{2} - \frac{1}{5} \right) \\
 &= \frac{5}{2} - 1 = \frac{3}{2} \rightarrow (3)
 \end{aligned}$$

From equation (2) and (3)

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

Hence Green's Theorem is verified.

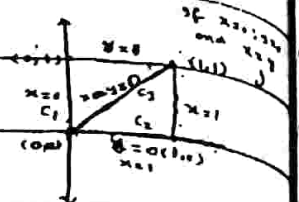
⑥ Given that

$$\oint Mdx + Ndy = \oint_C (3x^2 - 8y^2)dx + (4y - 12xy)dy \rightarrow (3)$$

As, $x=0$; $y=0$; $x+y=1$

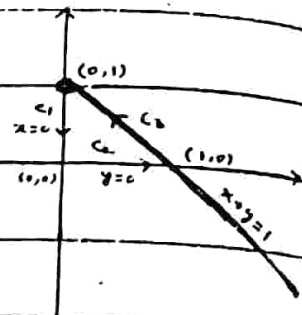
Along $x=0$:-

$$\begin{aligned} \int_{C_1} Mdx + Ndy &= \int_{y=0}^1 [0 + (4y+0)dy] \\ &= \int_{y=0}^1 4y dy \\ &= 2y^2 \Big|_0^1 \\ &= 2 \end{aligned}$$



Along $y=0$:-

$$\begin{aligned} \int_{C_2} Mdx + Ndy &= \int_{x=0}^1 (3x^2 - 0)dx \\ &= \int_{x=0}^1 3x^2 dx \\ &= x^3 \Big|_0^1 \\ &= 1 \end{aligned}$$



Along $(x+y=1)$

$y = -x+1 \Rightarrow dy = -dx$

$$\begin{aligned} \int_{C_3} Mdx + Ndy &= - \int_{x=0}^1 [(3x^2 - 8(-x+1)^2)dx + (4(-x+1) - 12x(-x+1))(-dx)] \\ &= - \int_{x=0}^1 (3x^2 - 8x^2 - 16x + 8 + 12x^2 - 4x + 12x^2 - 12x) dx \\ &= - \int_{x=0}^1 (-11x^2 + 26x - 12) dx \\ &= - \left(-\frac{11}{3}x^3 + 13x^2 - 12x \right) \Big|_0^1 \\ &= - \left(-\frac{11}{3} + 13 - 12 \right) \\ &= - \left(-\frac{11}{3} + 1 \right) \\ &= - \left(-\frac{8}{3} \right) = \frac{8}{3} \end{aligned}$$

The equation (3) becomes

$$\begin{aligned} \oint_C Mdx + Ndy &= \int_{C_1} Mdx + Ndy + \int_{C_2} Mdx + Ndy + \int_{C_3} Mdx + Ndy \\ &= \frac{8}{3} - 2 + 1 \\ &= \frac{5}{3} \rightarrow (2) \end{aligned}$$

$$M(x, y) = 3x^2 - 8y^2 \quad ; \quad N(x, y) = 4y - 6xy$$

$$\frac{\partial M}{\partial y} = -16y \quad ; \quad \frac{\partial N}{\partial x} = -6y$$

$$\begin{aligned} \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy &= \int_{x=0}^1 \int_{y=0}^{1-x} (-6y + 16y) dy dx \\ &= \int_{x=0}^1 \int_{y=0}^{1-x} (10y) dy dx \\ &= \int_{x=0}^1 \left(5y^2 \right)_{y=0}^{1-x} dx \\ &= \int_{x=0}^1 5(1-x)^2 dx \\ &= \int_{x=0}^1 5(1+x^2-2x) dx \\ &= 5 \left(x + \frac{x^3}{3} - 2x^2 \right) \Big|_0^1 \\ &= 5 \left(1 + \frac{1}{3} - 1 \right) \\ &= \frac{5}{3} \quad \text{--- (3)} \end{aligned}$$

From equation (2) and (3)

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

Hence Green's Theorem is Verified.

(38) Evaluate $\oint_C (3x+4y) dx + (2x-3y) dy$ where C, a circle of Radius 2 with Center at origin of the xy-plane, is traversed in positive sense.

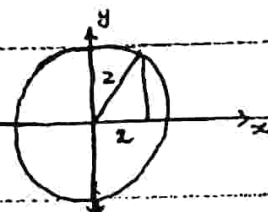
Sol.,

Given That

$$\oint_C M dx + N dy = \oint_C (3x+4y) dx + (2x-3y) dy \quad \text{--- (1)}$$

$$\text{Let } x = 2 \cos \theta \quad ; \quad y = 2 \sin \theta$$

$$dx = -2 \sin \theta d\theta \quad ; \quad dy = 2 \cos \theta d\theta \quad , \quad 0 \leq \theta < 2\pi$$



$$\begin{aligned}
 \oint Mdx + Ndy &= \int_0^{2\pi} (-8\sin\theta + 24(1 + \cos 2\theta)) d\theta - 48 \int_0^{2\pi} \cos^4 \theta d\theta \\
 &= \left[8\cos\theta + 24\left(\theta + \frac{\sin 2\theta}{2}\right) \right]_0^{2\pi} - 48 \times 4 \int_0^{\pi/2} \cos^4 \theta d\theta \\
 &= 8(0-0) + 24(2\pi + 0 - 0 - 0) - 48 \times 4 \cdot \frac{(4-1)(4-3)}{4(4-2)} \cdot \frac{\pi}{2} \\
 &= 48\pi - \frac{48 \times 4 \cdot 3 \cdot 1 \cdot \pi}{4 \cdot 2 \cdot 2} \\
 &= 48\pi - 36\pi \\
 &= 12\pi
 \end{aligned}$$

40) Evaluate $\oint (x^2 - 2xy)dx + (x^2y + 3)dy$ around the boundary of the region defined by $y^2 = 8x$ and $x = 2$.

a) directly b) by using Green's Theorem

Sol:-

a) Given that

$$\oint_C Mdx + Ndy = \oint_C (x^2 - 2xy)dx + (x^2y + 3)dy \rightarrow (1)$$

Along $\overline{DA}$

$$y = -\sqrt{8x} \quad ; \quad 0 \leq x \leq 2$$

$$dy = -\frac{\sqrt{2}}{\sqrt{x}} dx$$

The equation (1) becomes

$$\oint_{C_1} Mdx + Ndy = \int_{x=0}^2 (x^2 - 2x(-\sqrt{8x}))dx + (-x^2\sqrt{8x} + 3) \left(-\frac{\sqrt{2}}{\sqrt{x}} dx\right)$$

$$= \int_0^2 [x^2 + 4\sqrt{2}x^{3/2} + 4x^2 - 3\sqrt{2}x^{1/2}] dx$$

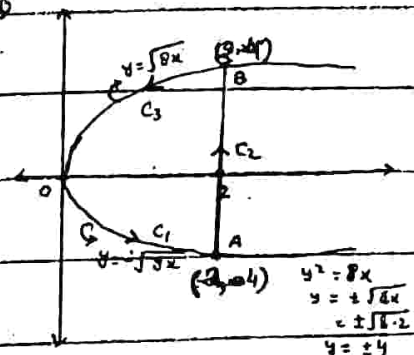
$$= \int_0^2 [5x^2 + 4\sqrt{2}x^{3/2} - 3\sqrt{2}x^{1/2}] dx$$

$$= \left[\frac{5}{3}x^3 + \frac{4\sqrt{2}}{5/2}x^{5/2} - \frac{3\sqrt{2}}{3/2}x^{3/2} \right]_{x=0}^2$$

$$= \frac{5}{3}(8) + \frac{8\sqrt{2}}{5}(2)^{5/2} - 6\sqrt{2} \cdot \sqrt{2}$$

$$= \frac{40}{3} + \frac{8\sqrt{2} \cdot 4\sqrt{2}}{5} - 12 = \frac{40}{3} + \frac{64}{5} - 12$$

$$= \frac{200 + 192 - 180}{15} = \frac{212}{15}$$



Along $\overline{AB}$.

$$x=2 \quad ; \quad dx=0 \quad ; \quad -4 \leq y \leq 4$$

The equation (2) becomes

$$\begin{aligned} \int_{C_2} Mdx + Ndy &= \int_{y=-4}^4 (2^2 y + 3) dy \\ &= \int_{y=-4}^4 (4y + 3) dy \\ &= \left(2y^2 + 3y \right) \Big|_{y=-4}^4 \\ &= 32 + 12 - 32 + 12 \\ &= 24 \end{aligned}$$

Along $\overline{BO}$.

$$y = \sqrt{8x} \quad ; \quad 2 \leq x \leq 0$$

$$dy = \frac{\sqrt{2}}{\sqrt{x}} dx$$

The equation (2) becomes

$$\begin{aligned} \int_{C_3} Mdx + Ndy &= \int_{x=2}^0 (x^2 - 2x\sqrt{8x}) dx + (x^2\sqrt{8x} + 3) \left(\frac{\sqrt{2}}{\sqrt{x}} dx \right) \\ &= \int_{x=2}^0 (x^2 - 2x\sqrt{8x} + 4x^2 + 3\sqrt{2}x^{1/2}) dx \\ &= - \int_{x=0}^2 (5x^2 - 4\sqrt{2}x^{3/2} + 3\sqrt{2}x^{1/2}) dx \\ &= - \left(\frac{5}{3}x^3 - \frac{4\sqrt{2}}{5/2}x^{5/2} + 3\sqrt{2}x^{3/2} \right) \Big|_{x=0}^2 \\ &= - \left(\frac{5}{3}(8) - \frac{8\sqrt{2}}{5}2^{5/2} + 6\sqrt{2} \cdot \sqrt{2} \right) \\ &= - \left(\frac{40}{3} - \frac{8\sqrt{2} \cdot 4\sqrt{2}}{5} + 12 \right) \\ &= - \left(\frac{40}{3} - \frac{64}{5} + 12 \right) \\ &= - \left(\frac{200 - 192 + 180}{15} \right) \\ &= - \frac{188}{15} \end{aligned}$$

As,

$$\begin{aligned} \oint_C Mdx + Ndy &= \int_{C_1} Mdx + Ndy + \int_{C_2} Mdx + Ndy + \int_{C_3} Mdx + Ndy \\ &= \frac{212}{15} - \frac{188}{15} + 24 \\ &= \frac{128}{5} \end{aligned}$$

$$M(x, y) = x^2 - 2xy$$

$$N(x, y) = x^2y + y$$

$$\frac{\partial M}{\partial y} = -2x$$

$$\frac{\partial N}{\partial x} = 2xy$$

$$0 \leq x \leq 2 \quad \text{and} \quad -\sqrt{8x} \leq y \leq \sqrt{8x}$$

$$\begin{aligned} \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy &= \int_{x=0}^2 \int_{y=-\sqrt{8x}}^{\sqrt{8x}} (2xy + 2x) dy dx \\ &= \int_{x=0}^2 \left[xy^2 + 2xy \right]_{y=-\sqrt{8x}}^{\sqrt{8x}} dx \\ &= \int_{x=0}^2 \left[2x \left(\frac{y^2 + y}{2} \right) \right]_{y=-\sqrt{8x}}^{\sqrt{8x}} dx \\ &= \int_{x=0}^2 2x \left(\frac{\sqrt{8x} + \sqrt{8x}}{2} \right) dx \\ &= \int_{x=0}^2 2x (2\sqrt{2x}) dx \\ &= \int_{x=0}^2 4x\sqrt{2x} dx \\ &= \int_{x=0}^2 4x \cdot 2\sqrt{2} \sqrt{x} dx \\ &= \int_{x=0}^2 8\sqrt{2} \cdot x^{3/2} dx \\ &= \left(8\sqrt{2} \cdot \frac{x^{5/2}}{5/2} \right) \Big|_{x=0}^2 \\ &= \frac{16\sqrt{2}}{5} \cdot 2^{5/2} \\ &= \frac{16\sqrt{2} \cdot 4\sqrt{2}}{5} \\ &= \frac{64 \times 2}{5} \\ &= \frac{128}{5} \quad \text{--- (3)} \end{aligned}$$

From equation (2) and (3)

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

Hence The Green's Theorem is Verified

(41) Evaluate $\int_{(0,0)}^{(\pi,2)} (6xy - y^2) dx + (3x^2 - 2xy) dy$ along the cycloid

$$x = \theta - \sin \theta, \quad y = 1 - \cos \theta$$

Sol:-

Given That

$$\oint_C M dx + N dy = \int_{(0,0)}^{(\pi,2)} (6xy - y^2) dx + (3x^2 - 2xy) dy \quad \text{--- (1)}$$

$$\text{As, } M(x,y) = 6xy - y^2 \quad ; \quad N(x,y) = 3x^2 - 2xy$$

$$\frac{\partial M}{\partial y} = 6x - 2y \quad ; \quad \frac{\partial N}{\partial x} = 6x - 2y$$

Since

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

This shows that the given equation is Exact so integral is independent of the path followed.

Let us take a particular path the straight line segment from $(0,0)$ to $(\pi,0)$ and $(\pi,0)$ to $(\pi,2)$

The equation (1) becomes

$$\oint_C M dx + N dy = \int_{x=0}^{\pi} (0) dx + 0 + \int_{y=0}^2 0 + (3\pi^2 - 2\pi y) dy$$

$$= 0 + \int_{y=0}^2 (3\pi^2 - 2\pi y) dy$$

$$= (3\pi^2 y - \pi y^2) \Big|_0^2$$

$$= 6\pi^2 - 4\pi$$

Alternatively:- Since

$$\oint_C M dx + N dy = \int_{(0,0)}^{(\pi,2)} d(3x^2 y - xy^2)$$

$$= (3x^2 y - xy^2) \Big|_{(0,0)}^{(\pi,2)}$$

$$= (6\pi^2 - 4\pi)$$

OR By using the straight line from $(0,0)$ to $(\pi,2)$

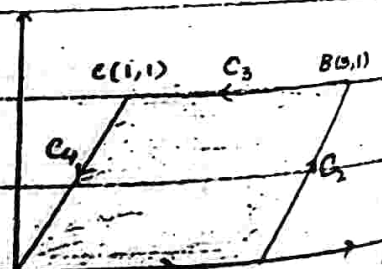
(42) Evaluate $\oint (3x^2 + 2y) dx - (x + 3\cos y) dy$ around the parallelogram having vertices at $(0,0)$, $(2,0)$, $(3,1)$ and $(1,1)$

Sol:-

Given That

$$\oint_C M dx + N dy = \oint_C (3x^2 + 2y) dx - (x + 3\cos y) dy \quad \text{--- (1)}$$

$$M(x,y) = 3x^2 + 2y \quad ; \quad N(x,y) = -x - 3\cos y$$



Along $\overline{OA}$ As, $0 \leq x \leq 2$, $y = 0$

The Equation (3) Becomes

$$\begin{aligned} \int_{C_1} Mdx + Ndy &= \int_0^2 \int_0^0 \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy \\ &= \int_0^2 \int_0^0 (2-1) dx (0) \\ &= 0 \end{aligned}$$

Along $\overline{AB}$ As, $2 \leq x \leq 3$ and $0 \leq y \leq 1$

The Equation (4) Becomes

$$\begin{aligned} \int_{C_2} Mdx + Ndy &= \int_2^3 \int_0^1 \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy \\ &= \int_2^3 \int_0^1 (-2-1) dx dy \\ &= \int_2^3 (-3)(1-0) dx \\ &= -3(3-2) \\ &= -3 \end{aligned}$$

Along $\overline{BC}$ As, $3 \leq x \leq 1$, $y = 1$

The equation (4) Becomes

$$\begin{aligned} \int_{C_3} Mdx + Ndy &= \int_3^1 \int_1^1 \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy \\ &= \int_3^1 \int_1^1 (-2-1) dx (0) \\ &= 0 \end{aligned}$$

Along $\overline{CD}$ As, $1 \leq x \leq 0$ and $1 \leq y \leq 0$

The equation (3) Becomes

$$\begin{aligned} \int_{C_4} Mdx + Ndy &= \int_1^0 \int_1^0 \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy \\ &= \int_1^0 \int_1^0 (-2-1) dx dy \end{aligned}$$

$$\int_{c_4} M dx + N dy = \int_{x=1}^0 -3(x-1) dx = \frac{-3(x^2-1)}{2} \Big|_1^0 = \frac{3(0-1)}{2} = -\frac{3}{2}$$

The equation (4) becomes

$$\oint_C M dx + N dy = \int_{c_1} M dx + N dy + \int_{c_2} M dx + N dy + \int_{c_3} M dx + N dy + \int_{c_4} M dx + N dy$$

$$= 0 - 3 + 0 - 3$$

$$= -6$$

Note: (Alternatively) we can evaluate the Question from straight lines.

(43) Find the Area bounded by one arch of the cycloid

$$x = a(\theta - \sin \theta), \quad y = a(1 - \cos \theta), \quad a > 0 \text{ and the } x\text{-axis}$$

Sol:-

Given that

$$x = a(\theta - \sin \theta), \quad y = a(1 - \cos \theta)$$

$$dx = a(1 - \cos \theta) d\theta; \quad dy = a \sin \theta d\theta$$

$$\text{where } 0 \leq \theta \leq 2\pi$$

The Area of simple closed curve 'C' is

$$\text{Area} = \frac{1}{2} \oint_C x dy - y dx$$

$$= \frac{1}{2} \int_{\theta=0}^{2\pi} a(\theta - \sin \theta) \cdot a \sin \theta d\theta - a(1 - \cos \theta) \cdot a(1 - \cos \theta) d\theta$$

$$= \frac{1}{2} \int_{\theta=0}^{2\pi} [a^2 \theta \sin \theta - a^2 \sin^2 \theta - a^2 (1 - \cos \theta)^2] d\theta$$

$$= \frac{a^2}{2} \int_{\theta=0}^{2\pi} [\theta \sin \theta - \sin^2 \theta - 1 + \cos^2 \theta + 2 \cos \theta] d\theta$$

$$= \frac{a^2}{2} \int_{\theta=0}^{2\pi} [\theta \sin \theta - \sin^2 \theta - 1 + 1 + \sin^2 \theta + 2 \cos \theta] d\theta$$

$$= \frac{a^2}{2} \int_{\theta=0}^{2\pi} [\theta \sin \theta - 2 + 2 \cos \theta] d\theta$$

$$= \frac{a^2}{2} \left[-\theta \cos \theta + \sin \theta - 2\theta + 2 \sin \theta \right]_{\theta=0}^{2\pi}$$

$$= \frac{a^2}{2} [-2\pi + 0 + 0 - 0 - 2(2\pi - 0)] = -4\pi a^2$$

$$\text{Area} = \frac{a^2}{2} [-3\pi]$$

$$= -3\pi a^2$$

$\therefore$ Area can never be Negative so,

$$\text{Area} = 3\pi a^2$$

(44) Find The Area bounded by Hypocycloid $x^{2/3} + y^{2/3} = a^{2/3}$, $a > 0$

Sol:-

Given That The equation of Hypocycloid are

$$x^{2/3} + y^{2/3} = a^{2/3} \quad a > 0$$

The parametric equations of Hypocycloid are

$$x = a \cos^3 \theta \quad ; \quad y = a \sin^3 \theta$$

$$dx = -3a \cos^2 \theta \sin \theta d\theta; \quad dy = 3a \sin^2 \theta \cos \theta d\theta$$

$$\text{where } 0 \leq \theta \leq 2\pi$$

The Area of simple closed curve is

$$\begin{aligned} \text{Area} &= \frac{1}{2} \oint_C x dy - y dx \\ &= \frac{1}{2} \int_{\theta=0}^{2\pi} a \cos^3 \theta (-3a \cos^2 \theta \sin \theta d\theta) + a \sin^3 \theta (3a \sin^2 \theta \cos \theta d\theta) \\ &= \frac{1}{2} \int_{\theta=0}^{2\pi} (-3a^2 \cos^5 \theta \sin \theta + 3a^2 \sin^5 \theta \cos \theta) d\theta \\ &= \frac{1}{2} \int_{\theta=0}^{2\pi} 3a^2 [\cos^4 \theta (-\sin \theta) + \sin^4 \theta (\cos \theta)] d\theta \\ &= \frac{1}{2} \int_{\theta=0}^{2\pi} 3a^2 [\sin^2 \theta \cos^2 \theta (-\sin^2 \theta + \cos^2 \theta)] d\theta \\ &= \frac{3a^2}{2} \int_{\theta=0}^{2\pi} (\sin^2 \theta \cos^2 \theta) d\theta \\ &= \frac{3a^2}{2} \cdot \frac{1}{4} \int_{\theta=0}^{2\pi} (4 \sin^2 \theta \cos^2 \theta) d\theta \\ &= \frac{3a^2}{2} \cdot \frac{1}{4} \int_{\theta=0}^{2\pi} (2 \sin \theta \cos \theta)^2 d\theta \\ &= \frac{3a^2}{2} \cdot \frac{1}{4} \int_{\theta=0}^{2\pi} \sin^2 2\theta d\theta \\ &= \frac{3a^2}{8} \int_{\theta=0}^{2\pi} \frac{1 - \cos 4\theta}{2} d\theta \end{aligned}$$

$$\text{Area} = \frac{3}{16} a^2 \left(\phi - \sin \frac{1}{2} \phi \right) \Big|_0^{2\pi}$$

$$= \frac{3}{16} a^2 (2\pi - 0 - 0 + 0)$$

$$= \frac{3}{16} a^2 (2\pi)$$

$$= \frac{3\pi a^2}{8}$$

(45) Show that in polar coordinates (r, θ) the expression

$$x dy - y dx = r^2 d\theta \quad \text{Interpret} \quad \frac{1}{2} \int x dy - y dx$$

Soln-

we know by the polar coordinates

$$x = r \cos \theta \quad ; \quad y = r \sin \theta$$

$$dx = dr \cos \theta - r \sin \theta d\theta \quad ; \quad dy = dr \sin \theta + r \cos \theta d\theta$$

Now

$$x dy - y dx = r \cos \theta [dr \sin \theta + r \cos \theta d\theta] - r \sin \theta [dr \cos \theta - r \sin \theta d\theta]$$

$$= r dr \sin \theta \cos \theta + r^2 \cos^2 \theta d\theta - r dr \sin \theta \cos \theta + r^2 \sin^2 \theta d\theta$$

$$= r^2 (\sin^2 \theta + \cos^2 \theta) d\theta$$

$$= r^2 (1) d\theta$$

$$= r^2 d\theta$$

By Green's Theorem in the plane

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

Put $M = -y$ and $N = x$

$$\oint_C -y dx + x dy = \iint_R \left(\frac{\partial x}{\partial x} - \frac{\partial (-y)}{\partial y} \right) dx dy$$

$$\oint_C x dy - y dx = \iint_R (1 + 1) dx dy$$

$$= 2 \iint_R dx dy$$

$$= 2 (\text{Area})$$

$$\text{Area} = \frac{1}{2} \oint_C x dy - y dx$$

This is the Area bounded by ~~Simple~~ Closed Curve C in Cartesian Coordinates

$$\text{Area} = \frac{1}{2} \oint_C r^2 d\theta$$

This is the Area bounded by Simple Closed Curve C in Polar Coordinates.

(46) Find the Area of a loop of the four-leaved rose

$$r = 3 \sin 2\theta$$

Sol:-

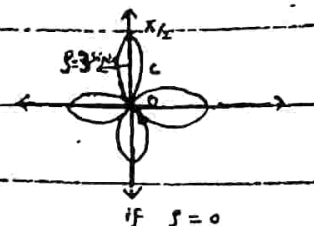
Given that

$$r = 3 \sin 2\theta$$

The Area of simple closed curve C

is given by

$$\begin{aligned} \text{Area} &= \frac{1}{2} \int_C r^2 d\theta \\ &= \frac{1}{2} \int_0^{\pi/2} (3 \sin 2\theta)^2 d\theta \\ &= \frac{1}{2} \int_0^{\pi/2} 9 \sin^2 2\theta d\theta \\ &= \frac{9}{2} \int_0^{\pi/2} \left(\frac{1 - \cos 4\theta}{2} \right) d\theta \\ &= \frac{9}{4} \left(\theta - \frac{\sin 4\theta}{4} \right) \Big|_0^{\pi/2} \\ &= \frac{9}{4} \left(\frac{\pi}{2} - 0 + 0 + 0 \right) \\ &= \frac{9\pi}{8} \end{aligned}$$



if $r = 0$

$$3 \sin 2\theta = 0$$

$$\sin 2\theta = 0$$

$$2\theta = \sin^{-1}(0)$$

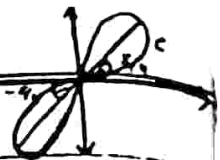
$$2\theta = 0, \pi$$

$$\theta = 0, \pi/2$$

(47) Find the Area of both loops of the lemniscate

$$r^2 = a^2 \cos 2\theta$$

Sol:-



Given that

$$r^2 = a^2 \cos 2\theta$$

if $r = a$

$$a^2 \cos 2\theta = 0$$

$$\cos 2\theta = 0$$

$$2\theta = \cos^{-1}(0)$$

$$2\theta = \pm \frac{\pi}{2}$$

$$\theta = \pm \frac{\pi}{4}$$

The Area of simple closed curve C

is given by

$$\text{Area} = \frac{1}{2} \oint_C r^2 d\theta$$

$$= \frac{1}{2} \int_{-\pi/4}^{\pi/4} a^2 \cos 2\theta d\theta$$

$$= \frac{1}{2} \int_{-\pi/4}^{\pi/4} a^2 \cos 2\theta d\theta$$

$$= \frac{1}{2} \int_{-\pi/4}^{\pi/4} a^2 \cos 2\theta d\theta$$

$$= \frac{1}{2} \cdot \frac{a^2}{2} \sin 2\theta \Big|_{-\pi/4}^{\pi/4}$$

$$= \frac{1}{2} \cdot \frac{a^2}{2} \left(\frac{1}{2} - (-\frac{1}{2}) \right)$$

$$= \frac{1}{2} \cdot \frac{a^2}{2} \left(\frac{1}{2} \right)$$

$$= \frac{1}{2} \cdot \frac{a^2}{2}$$

$$= \frac{1}{2} \cdot \frac{a^2}{2}$$

(48) Find the Area of the loop of the folium of Descartes $x^3 + y^3 = 3axy$, $a > 0$

Descartes $x^3 + y^3 = 3axy$, $a > 0$

Sol:

Given that

$$x^3 + y^3 = 3axy \quad \text{--- (1)}$$

$$\text{Let } y = xt$$

then for $x = 0, \frac{3}{2}a$ and $y = 0, \frac{3}{2}a$ gives

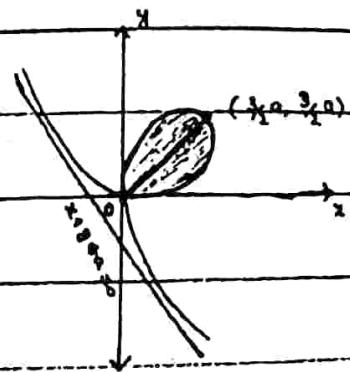
$$0 \leq t \leq 1$$

The equation (1) becomes

$$x^3 + x^3 t^3 = 3ax \cdot xt$$

$$x^3 (1 + t^3) = 3ax^2 t$$

$$x(1 + t^3) = 3at$$



$$x = \frac{3at}{1+t^2}$$

The Area of the simple closed curve C is

$$A = \text{Area} = \frac{1}{2} \oint_C x dy - y dx$$

$$A = \frac{1}{2} \oint_C x^2 \left(\frac{x dy - y dx}{x^2} \right)$$

$$= \frac{1}{2} \oint_C x^2 d\left(\frac{y}{x}\right)$$

$$= \frac{1}{2} \int_{t=0}^1 \frac{(3at)^2}{(1+t^2)^2} dt$$

$$= \frac{1}{2} \int_{t=0}^1 (1+t^2)^{-2} (9a^2 t^2) dt$$

$$= \frac{3a^2}{2} \int_{t=0}^1 (1+t^2)^{-2} (3t^2) dt$$

$$= \frac{3a^2}{2} \left[\frac{(1+t^2)^{-1}}{-1} \right]_0^1$$

$$= -\frac{3a^2}{2} \left[(1+1)^{-1} - (1+0)^{-1} \right]$$

$$= -\frac{3a^2}{2} \left[\frac{1}{2} - 1 \right]$$

$$= -\frac{3a^2}{2} \left[\frac{1}{2} - 1 \right]$$

$$= -\frac{3a^2}{2} \left[-\frac{1}{2} \right]$$

$$A = \frac{3a^2}{4}$$

$$\text{Required Area} = 2 \times \frac{3a^2}{4} = \frac{3a^2}{2}$$

* * *

(49) Verify Green's Theorem in the plane for $\oint_C (2x - y^3) dx - xy dy$,

where C is the boundary of the region enclosed by the

Circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 9$

Sol:-

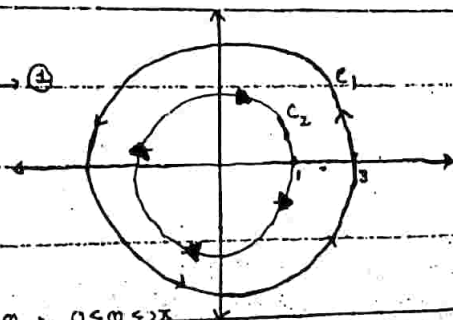
Given that

$$\oint_C M dx + N dy = \oint_C (2x - y^3) dx - xy dy \rightarrow \text{①}$$

For C_1 :-

$$\text{where } x = 3 \cos \theta, \quad y = 3 \sin \theta$$

$$dx = -3 \sin \theta d\theta, \quad dy = 3 \cos \theta d\theta; \quad 0 \leq \theta < 2\pi$$



The equation (2) becomes

$$\oint_C Mdx + Ndy = \int_{C_1} Mdx + Ndy + \int_{C_2} Mdx + Ndy$$

$$= \frac{243\pi}{4} - \frac{3\pi}{4}$$

$$= \frac{240\pi}{4}$$

$$= 60\pi \quad \rightarrow (2)$$

Now

$$M(x, y) = 2x - y^3 \quad ; \quad N = -xy$$

$$\frac{\partial M}{\partial y} = -3y^2 \quad ; \quad \frac{\partial N}{\partial x} = -y$$

By Green's Theorem in the plane

$$\oint_C Mdx + Ndy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy = \iint_R (-y + 3y^2) dx dy \rightarrow (3)$$

$$\text{Let } x = r \cos \theta \quad ; \quad y = r \sin \theta$$

$$\text{As } dx dy = r dr d\theta \quad ; \quad r^2 = x^2 + y^2$$

$$\text{Where } 1 \leq r \leq 3 \quad \text{and } 0 \leq \theta \leq 2\pi$$

The Equation (3) becomes

$$\iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy = \int_{\theta=0}^{2\pi} \int_{r=1}^3 (-r \sin \theta + 3r^2 \sin^2 \theta) r dr d\theta$$

$$= \int_{\theta=0}^{2\pi} \int_{r=1}^3 (-r^2 \sin \theta + 3r^3 \sin^2 \theta) dr d\theta$$

$$= \int_{\theta=0}^{2\pi} \left(-\frac{r^3}{3} \sin \theta + \frac{3}{4} r^4 \sin^2 \theta \right) \Big|_{r=1}^3 d\theta$$

$$= \int_{\theta=0}^{2\pi} \left(\left(-\frac{27}{3} + \frac{1}{3} \right) \sin \theta + \frac{3}{4} (81 - 1) \sin^2 \theta \right) d\theta$$

$$= \int_{\theta=0}^{2\pi} \left(-\frac{26}{3} \sin \theta + 60 \sin^2 \theta \right) d\theta$$

$$= \int_{\theta=0}^{2\pi} \left(-\frac{26}{3} \sin \theta + 30 (1 - \cos 2\theta) \right) d\theta$$

$$= \left(\frac{26}{3} \cos \theta + 30 \left(\theta - \frac{\sin 2\theta}{2} \right) \right) \Big|_{\theta=0}^{2\pi}$$

$$= \frac{26}{3} (1 - 1) + 30 (2\pi - 0 - 0 + 0)$$

$$= 0 + 60\pi$$

$$= 60\pi \rightarrow (4)$$

From equation (2) and (3)

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial M}{\partial x} - \frac{\partial N}{\partial y} \right) dx dy$$

Hence Green's Theorem is verified

(5) Evaluate $\int_{(1,0)}^{(-1,0)} \frac{-y dx + x dy}{x^2 + y^2}$ along the following paths

(a) straight line segments from (1,0) to (1,1) then to (-1,1) then to (-1,0)

(b) straight line segments from (1,0) to (1,-1) then to (-1,-1) then to (-1,0)

Show that although $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, the line integral is dependent on the path joining (1,0) to (-1,0) and

Explain.

Soln.

(a) As,

$$\int_{(1,0)}^{(-1,0)} \frac{-y dx + x dy}{x^2 + y^2} = \int_{(1,0)}^{(1,1)} \frac{dy}{1+y^2} + \int_{(1,1)}^{(-1,1)} \frac{-dx}{1+x^2} + \int_{(-1,1)}^{(-1,0)} \frac{-dy}{1+y^2}$$

$$= \left[\tan^{-1} y \right]_{(1,0)}^{(1,1)} - \left[\tan^{-1} x \right]_{(1,1)}^{(-1,1)} - \left[\tan^{-1} y \right]_{(-1,1)}^{(-1,0)}$$

$$= \tan^{-1}(1) - \tan^{-1}(0) - \tan^{-1}(-1) + \tan^{-1}(1) - \tan^{-1}(0) - \tan^{-1}(1)$$

$$= \frac{\pi}{4} - 0 + \frac{\pi}{4} + \frac{\pi}{4} - 0 + \frac{\pi}{4}$$

$$= \pi$$

(b) As,

$$\int_{(1,0)}^{(-1,0)} \frac{-y dx + x dy}{x^2 + y^2} = \int_{(1,0)}^{(1,-1)} \frac{dy}{1+y^2} + \int_{(1,-1)}^{(-1,-1)} \frac{dx}{1+x^2} + \int_{(-1,-1)}^{(-1,0)} \frac{-dy}{1+y^2}$$

$$= \left[\tan^{-1} y \right]_{(1,0)}^{(1,-1)} + \left[\tan^{-1} x \right]_{(1,-1)}^{(-1,-1)} - \left[\tan^{-1} y \right]_{(-1,-1)}^{(-1,0)}$$

$$= \tan^{-1}(-1) + \tan^{-1}(-1) - \tan^{-1}(0) - \tan^{-1}(0) + \tan^{-1}(1)$$

$$= -\frac{\pi}{4} - \frac{\pi}{4} - 0 - 0 + \frac{\pi}{4}$$

$$= -\frac{\pi}{4}$$

Since

$$\oint_C M dx + N dy = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \right) dx dy$$

where

$$M(x, y) = -\frac{y}{x^2 + y^2}$$

$$\frac{\partial M}{\partial y} = \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

$$\frac{\partial N}{\partial x} = \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

As

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

This shows that although $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$, the line integral

is dependent on the path joining $(1, 0)$ to $(1, 0)$ which

is clear from (a) and (b) because $M = -\frac{y}{x^2 + y^2}$ and

$N = \frac{x}{x^2 + y^2}$ do not have continuous derivatives throughout

any region including $(0, 0)$.

(51) By changing variables from (x, y) to (u, v) according to the

transformation $x = x(u, v)$, $y = y(u, v)$, show that the area

A of a region R bounded by a simple closed C is given by

$$A = \iint_R |J\left(\frac{x, y}{u, v}\right)| du dv \quad \text{where} \quad J\left(\frac{x, y}{u, v}\right) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

is the Jacobian of x and y with respect to u and v .

What restrictions should you make? Illustrate the result

where u and v are polar coordinates.

Soln.

(i) Let $\vec{r} = x\hat{i} + y\hat{j}$

$$\vec{r} = x(u,v)\hat{i} + y(u,v)\hat{j}$$

As $\vec{r}_u = \frac{\partial x}{\partial u}\hat{i} + \frac{\partial y}{\partial u}\hat{j}$

$$\vec{r}_v = \frac{\partial x}{\partial v}\hat{i} + \frac{\partial y}{\partial v}\hat{j}$$

Now $\vec{r}_u \times \vec{r}_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & 0 \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & 0 \end{vmatrix}$

$$\vec{r}_u \times \vec{r}_v = \left(\frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u} \right) \hat{k}$$

We know

$$\text{Area} = \frac{1}{2} \oint_C x dy - y dx$$

By Green's theorem

$$= \frac{1}{2} \iint_R \left(2(x) - 2(-y) \right) dx dy$$

$$= \frac{1}{2} \iint_R 2x dx dy$$

Directly From

the Surface Area

$$\iint_S ds = \iint_R |\vec{r}_u \times \vec{r}_v| du dv$$

$$= \iint_R J\left(\frac{x,y}{u,v}\right) du dv$$

$$\text{Area} = \iint_R dx dy$$

$$= \iint_R (\vec{r}_u \times \vec{r}_v) \cdot \hat{k} du dv$$

$$= \iint_R \left(\frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u} \right) (\hat{k} \cdot \hat{k}) du dv$$

$$= \iint_R \left(\frac{\partial x}{\partial u} \frac{\partial y}{\partial v} - \frac{\partial x}{\partial v} \frac{\partial y}{\partial u} \right) du dv$$

$$= \iint_R J\left(\frac{x,y}{u,v}\right) du dv$$



$$\begin{aligned} d\vec{S} &= (\vec{r}_u \times \vec{r}_v) du dv \\ \hat{k} ds &= (\vec{r}_u \times \vec{r}_v) du dv \\ ds &= (\vec{r}_u \times \vec{r}_v) \cdot \hat{k} du dv \\ dx dy &= (\vec{r}_u \times \vec{r}_v) \cdot \hat{k} du dv \end{aligned}$$

(ii) A parameterized surface should be smooth and $\vec{r}_u, \vec{r}_v$

are continuous, $\vec{r}_u \times \vec{r}_v \neq 0$ in the parameter domain

(iii) For polar Coordinates Let $x = r \cos \theta$, $y = r \sin \theta$

So

$$J\left(\frac{x,y}{r,\theta}\right) = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r$$

where $dx dy = r dr d\theta$

$$\text{So, Area} = \iint_R J\left(\frac{x,y}{r,\theta}\right) dr d\theta = \iint_R r dr d\theta$$

Evaluate $\iint_S \vec{F} \cdot \vec{n} \, ds$ where $\vec{F} = (xyz)\hat{i} + yz^2\hat{j} + xz\hat{k}$ and S is

(a) The surface of the parallelepiped bounded by $x=0, y=0$

$z=0, x=2, y=1$ and $z=3$

(b) The surface of the region bounded by $x=0, y=0, y=3$

$z=0$ and $x+2z=6$

Soln

Given that

$$\vec{F} = xyz\hat{i} + yz^2\hat{j} + xz\hat{k}$$

$$\vec{\nabla} \cdot \vec{F} = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (xyz\hat{i} + yz^2\hat{j} + xz\hat{k})$$

$$= \frac{\partial}{\partial x}(xyz) + \frac{\partial}{\partial y}(yz^2) + \frac{\partial}{\partial z}(xz)$$

$$= yz + z^2 + x$$

$$= x + yz + z^2$$

(a) By Divergence Theorem

$$\iint_S \vec{F} \cdot \vec{n} \, ds = \iiint_V \vec{\nabla} \cdot \vec{A} \, dv$$

$$= \int_{x=0}^2 \int_{y=0}^1 \int_{z=0}^3 (x + yz + z^2) \, dz \, dy \, dx$$

$$= \int_{x=0}^2 \int_{y=0}^1 \left[(xz + yz^2 + \frac{z^3}{3}) \Big|_0^3 \right] \, dy \, dx$$

$$= \int_{x=0}^2 \int_{y=0}^1 (3x + 6y + 9) \, dy \, dx$$

$$= \int_{x=0}^2 \left[(3xy + 3y^2 + 9y) \Big|_0^1 \right] \, dx$$

$$= \int_{x=0}^2 (3x + 3 + 9) \, dx$$

$$= \int_{x=0}^2 (3x + 12) \, dx$$

$$= \left(\frac{3x^2}{2} + 12x \right) \Big|_0^2$$

$$= \frac{3 \cdot 2^2}{2} + 12 \cdot 2$$

$$= 6 + 24$$

$$= 30$$

(b) By Divergence Theorem

$$\iiint_V \vec{F} \cdot \vec{n} \, ds = \iiint_V \vec{\nabla} \cdot \vec{F} \, dv$$

$$= \int_{z=0}^3 \int_{y=0}^3 \int_{x=0}^{6-2z} (x+zy+z^2) \, dx \, dy \, dz$$

$$= \int_{z=0}^3 \int_{y=0}^3 \left[\frac{x^2}{2} + zxy + xz^2 \right]_0^{6-2z} dy \, dz$$

$$= \int_{z=0}^3 \int_{y=0}^3 \left[\frac{(6-2z)^2}{2} + 2y(6-2z) + (6-2z)z^2 \right] dy \, dz$$

$$= \int_{z=0}^3 \int_{y=0}^3 \left[\frac{(6-2z)^2}{2} + 2y(6-2z) + (6z^2 - 2z^3) \right] dy \, dz$$

$$= \int_{z=0}^3 \left[\frac{(6-2z)^2}{2} \cdot y + y^2(6-2z) + (6z^2 - 2z^3) \cdot y \right]_0^3 dz$$

$$= \int_{z=0}^3 \left[\frac{3}{2} (6-2z)^2 + 9(6-2z) + 3(6z^2 - 2z^3) \right] dz$$

$$= \int_{z=0}^3 \left[-\frac{3}{4} (6-2z)^2 (-2) - \frac{9}{2} (6-2z)(-2) + 18z^2 - 6z^3 \right] dz$$

$$= \left[-\frac{3}{4} \frac{(6-2z)^3}{3} - \frac{9}{2} \frac{(6-2z)^2}{2} + 6z^3 - \frac{6}{4} z^4 \right]_0^3$$

$$= -\frac{1}{4} (0 - 6^3) - \frac{9}{4} (-6^2) + 6 \cdot 3^3 - \frac{6}{4} \cdot 3^4$$

$$= -\frac{1}{4} (-216) - \frac{9}{4} (-36) + 162 - \frac{186}{4}$$

$$= \frac{216}{4} + \frac{324}{4} + \frac{162}{4} - \frac{186}{4}$$

$$= \frac{216 + 324 + 162 - 186}{4}$$

$$= \frac{1188 - 486}{4}$$

$$= \frac{702}{4}$$

$$= \frac{351}{2}$$

Alternatively,

we can consider the planes and curve

(53) verify the divergence theorem for $\vec{A} = 2x^2y\hat{i} - y^2\hat{j} + 4xz^2\hat{k}$ taken over the region bounded by $y^2 + z^2 = 9$ and $x=2$ in first octant.

Soln.

Given that

$$\vec{A} = 2x^2y\hat{i} - y^2\hat{j} + 4xz^2\hat{k}$$

$$y^2 + z^2 = 9 \text{ and } x=2$$

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$$\vec{\nabla} \cdot \vec{A} = \frac{\partial}{\partial x}(2x^2y) + \frac{\partial}{\partial y}(-y^2) + \frac{\partial}{\partial z}(4xz^2)$$

$$= 4xy - 2y + 8xz$$

Now

$$\iiint_V \vec{\nabla} \cdot \vec{A} \, dv = \iiint_V (4xy - 2y + 8xz) \, dv$$

$$= \int_{x=0}^2 \int_{y=0}^3 \int_{z=0}^{\sqrt{9-y^2}} (4xy - 2y + 8xz) \, dz \, dy \, dx$$

$$= \int_{x=0}^2 \int_{y=0}^3 \left[4xyz - 2yz + 4xz^2 \right]_{z=0}^{\sqrt{9-y^2}} dy \, dx$$

$$= \int_{x=0}^2 \int_{y=0}^3 [4xy\sqrt{9-y^2} - 2y\sqrt{9-y^2} + 4x(9-y^2)] dy \, dx$$

$$= \int_{x=0}^2 \int_{y=0}^3 \left[-2x \cdot \frac{(9-y^2)^{3/2}}{3/2} + \frac{(9-y^2)^{3/2}}{3/2} + 36x - 4xy^2 \right] dy \, dx$$

$$= \int_{x=0}^2 \left[-\frac{4x}{3} (0-27) + \frac{2}{3} (-27) + 108x - 36x \right] dx$$

$$= \int_{x=0}^2 (36x - 18 + 108x - 36x) dx$$

$$= \int_{x=0}^2 (108x - 18) dx$$

$$= \left(54x^2 - 18x \right) \Big|_{x=0}^2$$

$$= 54(4) - 18(2)$$

$$= 216 - 36$$

$$= 180 \quad \text{--- (4)}$$

As, For S_1 :-

$$\vec{A} = 2x^2y\hat{i} - y^2\hat{j} + 4xz^2\hat{k}$$

$$y^2 + z^2 = 9$$

$$\text{Let } \phi(x, y, z) = y^2 + z^2 - 9$$

$$\vec{\nabla} \phi = 2y\hat{j} + 2z\hat{k}$$

$$|\vec{\nabla} \phi| = \sqrt{4y^2 + 4z^2}$$

$$= \sqrt{4(y^2 + z^2)} = \sqrt{4(9)} = 6$$

$$\begin{aligned}\vec{n} &= \frac{\vec{\nabla}\phi}{|\vec{\nabla}\phi|} \\ &= \frac{2y\hat{j} + 2z\hat{k}}{6} \\ &= \frac{y}{3}\hat{j} + \frac{z}{3}\hat{k}\end{aligned}$$

$$|\vec{n}| = \frac{2}{3}$$

$$\begin{aligned}\vec{A} \cdot \vec{n} &= (2x^2y\hat{i} - y^2\hat{j} + 4xz^2\hat{k}) \cdot \left(\frac{y}{3}\hat{j} + \frac{z}{3}\hat{k}\right) \\ &= \frac{2x^2y^2(-1) - y^3 + 4xz^3}{3} \\ &= \frac{1}{3}[-y^3 + 4xz^3]\end{aligned}$$

$$z^2 = 9 - y^2 \quad \therefore z = \sqrt{9 - y^2}$$

$$\vec{A} \cdot \vec{n} = \frac{1}{3}[-y^3 + 4x(9 - y^2)\sqrt{9 - y^2}]$$

$$\begin{aligned}\text{Now } \iint_S \vec{A} \cdot \vec{n} \, dS &= \iint_R \frac{\vec{A} \cdot \vec{n}}{|\vec{n}|} \, dx \, dy \\ \iint_{S_1} \vec{A} \cdot \vec{n} \, dS &= \iint_R \left[\frac{1}{3}[-y^3 + 4x(9 - y^2)\sqrt{9 - y^2}] \right] \frac{dx \, dy}{\sqrt{9 - y^2}} \\ &= \iint_R \left[\frac{-y^3}{\sqrt{9 - y^2}} + 4x(9 - y^2) \right] dx \, dy \quad \text{--- (2)}\end{aligned}$$

$$y^2 + z^2 = 9$$

For y-intercept put $z=0$

$$y^2 = 9$$

$$y = \pm 3$$

As x varies from 0 to 2

y varies from -3 to 3

The equation (2) becomes

$$\begin{aligned}\iint_{S_1} \vec{A} \cdot \vec{n} \, dS &= \int_{y=-3}^3 \int_{x=0}^2 \left[\frac{-y^3}{\sqrt{9 - y^2}} + 4x(9 - y^2) \right] dx \, dy \\ &= \int_{y=-3}^3 \left[\frac{-2y^3}{\sqrt{9 - y^2}} + 8(9 - y^2) \right] dy \quad \text{--- (3)}\end{aligned}$$

Let $y = 3 \sin \alpha$

$$dy = 3 \cos \alpha \, d\alpha$$

where $0 \leq \theta \leq \pi/2$

The equation (3) becomes

$$\begin{aligned}
 \iint_{S_1} \vec{A} \cdot \vec{n} \, ds_1 &= \int_{\theta=0}^{\pi/2} \int_{\phi=0}^{2\pi} \left[\frac{-54 \sin^2 \theta}{3 \cos \theta} + 8(9 - 9 \sin^2 \theta) \right] 3 \cos \theta \, d\phi \, d\theta \\
 &= \int_{\theta=0}^{\pi/2} \int_{\phi=0}^{2\pi} [-54 \sin^2 \theta \sin \theta + 216(1 - \sin^2 \theta) \cos \theta] \, d\phi \, d\theta \\
 &= 54 \int_{\theta=0}^{\pi/2} \int_{\phi=0}^{2\pi} [(1 - \cos^2 \theta)(-\sin \theta) + 4 \cos \theta - 4 \sin^2 \theta \cos \theta] \, d\phi \, d\theta \\
 &= 54 \int_{\theta=0}^{\pi/2} \int_{\phi=0}^{2\pi} [-\sin \theta + 6 \cos \theta (-\sin \theta) + 4 \cos \theta - 4 \sin^2 \theta \cos \theta] \, d\phi \, d\theta \\
 &= 54 \int_{\theta=0}^{\pi/2} \left[\cos \theta - \cos^3 \theta + 4 \sin \theta - 4 \sin^3 \theta \right]_{\phi=0}^{2\pi} d\theta \\
 &= 54 \int_{\theta=0}^{\pi/2} \left[-1 + \frac{1}{3} + 4 - 4 \right] d\theta \\
 &= 54 \int_{\theta=0}^{\pi/2} [3 - 1] d\theta \\
 &= 54(2) \\
 &= 108
 \end{aligned}$$

For S_2 :

As, $x=2$; $\vec{n} = \hat{i}$

$$\begin{aligned}
 \vec{A} \cdot \vec{n} &= (2x^2y \hat{i} - y^2 \hat{j} + 4xz^2 \hat{k}) \cdot \hat{i} \\
 &= 2x^2y
 \end{aligned}$$

where $0 \leq y \leq 3$; $0 \leq z \leq \sqrt{9-y^2}$

So,

$$\begin{aligned}
 \iint_{S_2} \vec{A} \cdot \vec{n} \, ds_2 &= \int_{y=0}^3 \int_{z=0}^{\sqrt{9-y^2}} 2x^2y \, dz \, dy \\
 &= \int_{y=0}^3 (0y^2 \Big|_0^{\sqrt{9-y^2}}) dy \\
 &= \int_{y=0}^3 (0y \sqrt{9-y^2}) dy \\
 &= -24 \int_{y=0}^3 \sqrt{9-y^2} (-2y) dy \\
 &= -24 \cdot \frac{(9-y^2)^{3/2}}{3/2} \Big|_0^3 \\
 &= -\frac{8}{3} [0 - 27] \\
 &= -\frac{8}{3} [-27]
 \end{aligned}$$

Ans.

$$\iint_S \vec{A} \cdot \vec{n} \, ds = \iint_{S_1} \vec{A} \cdot \vec{n} \, ds_1 + \iint_{S_2} \vec{A} \cdot \vec{n} \, ds_2$$

$$= 108 + 72$$

$$= 180 \quad (4)$$

From equation (3) and (4)

$$\iiint_V \vec{\nabla} \cdot \vec{A} \, dv = \iint_S \vec{A} \cdot \vec{n} \, ds$$

Hence the Divergence Theorem is verified.

(S4) Evaluate $\iint_S \vec{r} \cdot \vec{n} \, ds$ where

(a) S is the sphere of radius 2 with center at (0,0,0)

(b) S is the surface of the cube bounded by $x=1, y=1, z=1$

$$x=-1, x=1, y=-1, y=1, z=-1, z=1$$

(c) S is the surface bounded by the paraboloid

$$z = 4 - (x^2 + y^2) \text{ and the } xy\text{-plane}$$

Sol:-

we know

$$\iint_S \vec{r} \cdot \vec{n} \, ds = 3V$$

(a) S is the surface of the sphere of radius 2 with

Equation:-

$$x^2 + y^2 + z^2 = 2^2$$

$$r^2 = 2^2$$

$$r = 2$$

$$\text{let } \vec{n} = \frac{\vec{r}}{r}$$

we

$$\iint_S \vec{r} \cdot \vec{n} \, ds = \iint_S \vec{r} \cdot \frac{\vec{r}}{r} \, ds$$

$$= 2 \iint_S ds$$

$$= 2 \cdot 4\pi(2)^2$$

$$= 2 \cdot 16\pi$$

$$= 32\pi$$

Center at (0,0,0) so,

$$x^2 + y^2 + z^2 = (2)^2$$

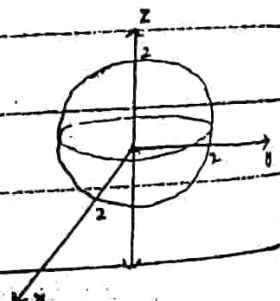
we know the volume of the sphere is

$$V = \frac{4}{3} \pi r^3 = \frac{4}{3} \pi (2)^3 = \frac{32\pi}{3}$$

So,

$$\iint_S \vec{r} \cdot \vec{n} \, ds = 3 \cdot \frac{32\pi}{3}$$

$$= 32\pi$$



② Surface S of the cube bounded by

$$x=-1, y=-1, z=-1, x=1, y=1, z=1$$

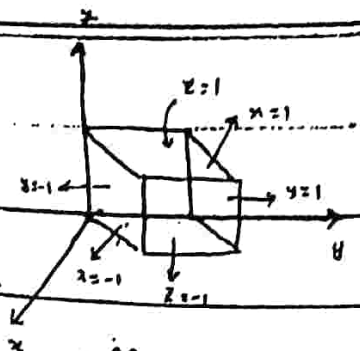
we know the volume of the cube is

$$V = a^3 = 2^3 = 8$$

So,

$$\iint_S \vec{r} \cdot \vec{n} \, ds = 3 \cdot 8$$

$$= 24$$



$$\begin{aligned} \text{or } \iiint_V dV &= \int_{-1}^1 \int_{-1}^1 \int_{-1}^1 dx \, dy \, dz \\ &= \int_{-1}^1 \int_{-1}^1 (2) \, dy \, dz \\ &= 8 \end{aligned}$$

③ S is the surface bounded by paraboloid

$$z = 4 - (x^2 + y^2) \text{ and the } xy\text{-plane.}$$

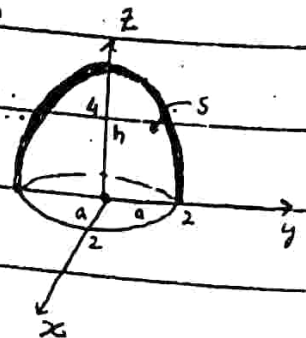
we know the volume of paraboloid is given by

$$V = \frac{1}{2} \pi a^2 h = \frac{1}{2} \pi \cdot 2^2 \cdot 4 = 8\pi$$

So,

$$\iint_S \vec{r} \cdot \vec{n} \, ds = 3 \cdot 8\pi$$

$$= 24\pi$$



⑤ If S is any closed surface enclosing a volume V and

$$\vec{A} = ax\hat{i} + by\hat{j} + cz\hat{k} \text{ Prove that}$$

$$\iint_S \vec{A} \cdot \vec{n} \, ds = (a+b+c)V$$

Sol:-

Given that

$$\vec{A} = ax\hat{i} + by\hat{j} + cz\hat{k}$$

$$\vec{\nabla} \cdot \vec{A} = \frac{\partial}{\partial x}(ax) + \frac{\partial}{\partial y}(by) + \frac{\partial}{\partial z}(cz)$$

$$= a+b+c$$

we know by Divergence Theorem

$$\begin{aligned}\iint_S \vec{A} \cdot \vec{n} \, ds &= \iiint_V (\vec{\nabla} \cdot \vec{A}) \, dv \\ &= \iiint_V (a+b+c) \, dv \\ &= (a+b+c) \iiint_V dv \\ &= (a+b+c) V\end{aligned}$$

(56) If $\vec{H} = \text{curl } \vec{A}$, Prove that $\iint_S \vec{H} \cdot \vec{n} \, ds = 0$ for any closed surface S .

Sol.

Given that

$$\vec{H} = \text{curl } \vec{A}$$

$$\vec{H} = \vec{\nabla} \times \vec{A}$$

we know by Divergence Theorem

$$\begin{aligned}\iint_S \vec{H} \cdot \vec{n} \, ds &= \iiint_V (\vec{\nabla} \cdot \vec{H}) \, dv \\ &= \iiint_V (\vec{\nabla} \cdot \vec{\nabla} \times \vec{A}) \, dv \\ &= 0\end{aligned}$$

(57) If $\vec{n}$ is the unit outward drawn to any closed surface of Area S , Show that

$$\iiint_V \text{div } \vec{A} \, dv = S$$

Sol.

Given that

$$\vec{A} = \vec{n}$$

we know by Divergence Theorem

$$\iiint_V \vec{\nabla} \cdot \vec{A} \, dv = \iint_S \vec{A} \cdot \vec{n} \, ds$$

$$\iiint_V \vec{\nabla} \cdot \vec{n} \, dv = \iint_S \vec{n} \cdot \vec{n} \, ds$$

$$\iiint_V \text{div } \vec{n} \, dv = \iint_S ds$$

$$= S$$

(58)

Prove

$$\iiint_V \frac{dv}{r^2} = \iint_S \frac{\vec{r} \cdot \vec{n}}{r^2} \, ds$$

Soln:

Let $\vec{A} = \frac{\vec{r}}{r^2}$

we know by Divergence Theorem

$$\iiint_V \vec{\nabla} \cdot \vec{A} \, dv = \iint_S \vec{A} \cdot \vec{n} \, ds$$

$$\iiint_V \vec{\nabla} \cdot \frac{\vec{r}}{r^2} \, dv = \iint_S \frac{\vec{r} \cdot \vec{n}}{r^2} \, ds$$

$$\vec{\nabla} \cdot \frac{\vec{r}}{r^2} = \left(\frac{\partial}{\partial x} \frac{x}{r^2} + \frac{\partial}{\partial y} \frac{y}{r^2} + \frac{\partial}{\partial z} \frac{z}{r^2} \right)$$

$$= \frac{\partial}{\partial x} \left(\frac{x}{x^2+y^2+z^2} \right) + \frac{\partial}{\partial y} \left(\frac{y}{x^2+y^2+z^2} \right) + \frac{\partial}{\partial z} \left(\frac{z}{x^2+y^2+z^2} \right)$$

$$= \frac{-x^2+y^2+z^2}{(x^2+y^2+z^2)^2} + \frac{-y^2+x^2+z^2}{(x^2+y^2+z^2)^2} + \frac{-z^2+x^2+y^2}{(x^2+y^2+z^2)^2}$$

$$= \frac{-x^2+y^2+z^2 - y^2+x^2+z^2 - z^2+x^2+y^2}{(x^2+y^2+z^2)^2}$$

$$= \frac{x^2+y^2+z^2 - x^2-y^2-z^2}{(x^2+y^2+z^2)^2}$$

$$= \frac{0}{(x^2+y^2+z^2)^2} = 0$$

Hence, $\iiint_V \frac{dv}{r^2} = \iint_S \frac{\vec{r} \cdot \vec{n}}{r^2} \, ds$

(59)

Prove

$$\iint_S r^5 \vec{n} \, ds = \iiint_V 5r^3 \vec{r} \, dv$$

Soln:

Let $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$$r = (x^2 + y^2 + z^2)^{1/2}$$

$$r^5 = (x^2 + y^2 + z^2)^{5/2}$$

$$\vec{\nabla} (r^5) = \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right) (x^2 + y^2 + z^2)^{5/2}$$

$$= 5(x^2 + y^2 + z^2)^{3/2} (2x\hat{i} + 2y\hat{j} + 2z\hat{k})$$

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$\vec{\nabla} r^5 = 5r^3 \vec{r}$

$$\begin{aligned}\vec{\nabla}(r^5) &= 5(\sqrt{x^2+y^2+z^2})^3(x\hat{i}+y\hat{j}+z\hat{k}) \\ &= 5r^3\vec{r}\end{aligned}$$

we know by Divergence Theorem

$$\begin{aligned}\iint_S r^5 \vec{n} \, ds &= \iiint_V \vec{\nabla}(r^5) \, dv \\ &= \iiint_V 5r^3 \vec{r} \, dv\end{aligned}$$

(60) Prove $\iint_S \vec{n} \, ds = 0$ for any closed surface S .

Proof.

we know By Gradient Theorem

$$\iint_S \phi \vec{n} \, ds = \iiint_V \nabla \phi \, dv$$

Let $\phi = 1$

$$\begin{aligned}\iint_S \vec{n} \, ds &= \iiint_V \vec{\nabla}(1) \, dv \\ &= 0\end{aligned}$$

(61) Show that Green's second identity can be written as

$$\iiint_V (\phi \nabla^2 \psi - \psi \nabla^2 \phi) \, dv = \iint_S \left(\phi \frac{d\psi}{dn} - \psi \frac{d\phi}{dn} \right) ds$$

Sol.

The Green's second identity is

$$\begin{aligned}\iiint_V (\phi \nabla^2 \psi - \psi \nabla^2 \phi) \, dv &= \iint_S (\phi \nabla \psi - \psi \nabla \phi) \cdot d\vec{s} \\ &= \iint_S (\phi \nabla \psi - \psi \nabla \phi) \cdot \vec{n} \, ds \\ &= \iint_S (\phi \nabla \psi \cdot \vec{n} - \psi \nabla \phi \cdot \vec{n}) \, ds\end{aligned}$$

$$\begin{aligned}\psi &= \frac{\partial \psi}{\partial n} \vec{n} \\ \nabla \psi \cdot \vec{n} &= \frac{\partial \psi}{\partial n} \\ \text{Similarly} \\ \nabla \phi \cdot \vec{n} &= \frac{\partial \phi}{\partial n}\end{aligned}$$

we know

$$\vec{\nabla} \psi \cdot \vec{n} = \frac{\partial \psi}{\partial n} = \frac{d\psi}{dn}, \quad \vec{\nabla} \phi \cdot \vec{n} = \frac{\partial \phi}{\partial n} = \frac{d\phi}{dn}$$

$$\iiint_V (\phi \nabla^2 \psi - \psi \nabla^2 \phi) \, dv = \iint_S \left(\phi \frac{d\psi}{dn} - \psi \frac{d\phi}{dn} \right) ds$$

(62) Prove $\oint_S \vec{r} \times d\vec{S} = 0$ for any closed surface S .

Sol:-

We know by the Curl Theorem

$$\oint_S \vec{n} \times \vec{B} \, ds = \iiint_V \vec{\nabla} \times \vec{B} \, dv$$

As,

$$\oint_S \vec{r} \times d\vec{S} = \oint_S \vec{r} \times \vec{n} \, ds$$

$$= - \oint_S \vec{n} \times \vec{r} \, ds$$

$$= - \iiint_V \vec{\nabla} \times \vec{r} \, dv$$

$$= 0 \quad \because \vec{\nabla} \times \vec{r} = 0$$

(63) Verify Stokes' Theorem for $\vec{A} = (y-z+2)\hat{i} + (yz+4)\hat{j} - xz\hat{k}$ where S is the surface of the cube $x=0, y=0, z=0, x=2, y=2, z=2$ above the xy -plane.

Sol:-

Given that

$$\vec{A} = (y-z+2)\hat{i} + (yz+4)\hat{j} - xz\hat{k}$$

As,

$$\oint_C \vec{A} \cdot d\vec{r} = \int_{O G F A O} (y-z+2)dx + (yz+4)dy - xzdz \quad \rightarrow (1)$$

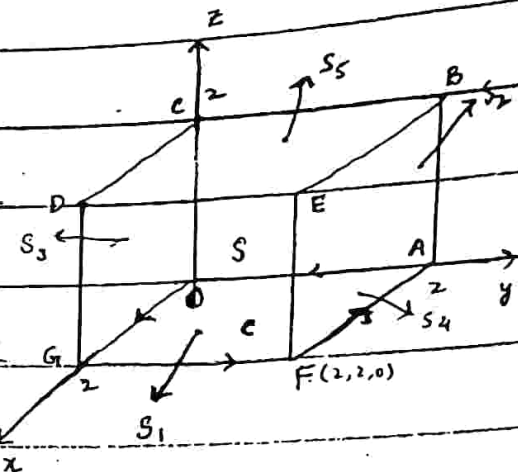
For $O G$:- As, $0 \leq x \leq 2, dy=dz=0, y=z=0$

$$\text{From (1)} \quad \int_{O G} \vec{A} \cdot d\vec{r} = \int_{x=0}^2 2 \, dx = 4$$

For $G F$:-

As, $x=2, z=0, dx=dz=0, 0 \leq y \leq 2$

$$\text{From (1)} \quad \int_{G F} \vec{A} \cdot d\vec{r} = \int_{y=0}^2 4 \, dy = 8$$



(*) Alternatively by taking

For $\vec{F}_A$:-

$$\text{As, } y=2, z=0 : dy=dz=0 : 2 \leq x \leq 0$$

The equation (1) becomes

$$\int_{FA} \vec{A} \cdot d\vec{r} = \int_{x=2}^0 4 dx = -8$$

For $\vec{A}_0$:-

$$\text{As, } x=0, z=0 : dx=dz=0 : 2 \leq y \leq 0$$

The equation (2) becomes

$$\int_{A_0} \vec{A} \cdot d\vec{r} = \int_{y=2}^0 4 dy = -8$$

The equation (3) becomes

$$\begin{aligned} \oint_C \vec{A} \cdot d\vec{r} &= \int_{OA} \vec{A} \cdot d\vec{r} + \int_{AF} \vec{A} \cdot d\vec{r} + \int_{FA} \vec{A} \cdot d\vec{r} + \int_{A_0} \vec{A} \cdot d\vec{r} \\ &= 4 + 8 - 8 - 8 \\ &= -4 \quad \text{--- (2)} \end{aligned}$$

Now

$$\vec{A} = (y-z+2)\hat{i} + (yz+4)\hat{j} - xz\hat{k}$$

$$\vec{\nabla} \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y-z+2 & yz+4 & -xz \end{vmatrix}$$

$$\vec{\nabla} \times \vec{A} = -y\hat{i} + (-1+z)\hat{j} - \hat{k}$$

$$\iint_S (\vec{\nabla} \times \vec{A}) \cdot \vec{n} \, dS = \iint_S [-y\hat{i} + (z-1)\hat{j} - \hat{k}] \cdot \vec{n} \, dS \quad \text{--- (3)}$$

For S_1 :-

$$\text{As, } x=2 : \vec{n} = \hat{i} : 0 \leq y \leq 2 : 0 \leq z \leq 2$$

The equation (3) becomes

$$\begin{aligned} \iint_{S_1} (\vec{\nabla} \times \vec{A}) \cdot \vec{n} \, dS &= \int_{y=0}^2 \int_{z=0}^2 -y \, dz \, dy \\ &= \int_{y=0}^2 -2y \, dy \\ &= -4 \end{aligned}$$

For S_2 :-

$$\text{As } z=0; \quad \vec{n} = -\hat{j}; \quad 0 \leq y \leq 2; \quad 0 \leq x \leq 2$$

The equation (3) becomes

$$\begin{aligned} \iint_{S_2} (\vec{\nabla} \times \vec{A}) \cdot \vec{n} \, ds_2 &= \int_{y=0}^2 \int_{z=0}^2 y \, dz \, dy \\ &= \int_{y=0}^2 2y \, dy \\ &= 4 \end{aligned}$$

For S_3 :-

$$\text{As } y=2; \quad \vec{n} = \hat{j}; \quad 0 \leq x \leq 2; \quad 0 \leq z \leq 2$$

The Equation (3) Becomes

$$\begin{aligned} \iint_{S_3} (\vec{\nabla} \times \vec{A}) \cdot \vec{n} \, ds_3 &= \int_{x=0}^2 \int_{z=0}^2 (z-1) \, dz \, dx \\ &= \int_{x=0}^2 \left(\frac{z^2}{2} - z \right) \Big|_0^2 \, dx \\ &= \int_{x=0}^2 (2-2) \, dx = 0 \end{aligned}$$

For S_4 :-

$$\text{As } y=0; \quad \vec{n} = -\hat{j}; \quad 0 \leq x \leq 2; \quad 0 \leq z \leq 2$$

The Equation (3) Becomes

$$\begin{aligned} \iint_{S_4} (\vec{\nabla} \times \vec{A}) \cdot \vec{n} \, ds_4 &= \int_{x=0}^2 \int_{z=0}^2 -(z-1) \, dz \, dx \\ &= \int_{x=0}^2 -(2-2) \, dx \\ &= 0 \end{aligned}$$

For S_5 :-

$$\text{As } x=2; \quad \vec{n} = \hat{i}; \quad 0 \leq x \leq 2; \quad 0 \leq y \leq 2$$

The Equation (3) Becomes

$$\begin{aligned} \iint_{S_5} (\vec{\nabla} \times \vec{A}) \cdot \vec{n} \, ds_5 &= \int_{x=0}^2 \int_{y=0}^2 -1 \, dy \, dx \\ &= \int_{x=0}^2 -2 \, dx \\ &= -4 \end{aligned}$$

As,

$$\begin{aligned}
 \iint_S (\nabla \times \vec{A}) \cdot \vec{n} \, ds &= \iint_{S_1} (\nabla \times \vec{A}) \cdot \vec{n} \, ds_1 + \iint_{S_2} (\nabla \times \vec{A}) \cdot \vec{n} \, ds_2 + \iint_{S_3} (\nabla \times \vec{A}) \cdot \vec{n} \, ds_3 \\
 &\quad + \iint_{S_4} (\nabla \times \vec{A}) \cdot \vec{n} \, ds_4 + \iint_{S_5} (\nabla \times \vec{A}) \cdot \vec{n} \, ds_5 \\
 &= -4 + 4 + 0 + 0 - 4 \\
 &= -4 \rightarrow \textcircled{4}
 \end{aligned}$$

From equation (2) and (4)

$$\oint_C \vec{A} \cdot d\vec{r} = \iint_S (\nabla \times \vec{A}) \cdot \vec{n} \, ds$$

Hence the Stokes' Theorem is verified.

- (64) Verify Stokes' Theorem for $\vec{F} = xz\hat{i} - y\hat{j} + x^2y\hat{k}$ where S is the surface of the region bounded by $x=4, y=4, z=4$ and $2x+y+2z=8$ which is not included in xz -plane.

Sol:-

Given that

$$\vec{F} = xz\hat{i} - y\hat{j} + x^2y\hat{k}$$

As,

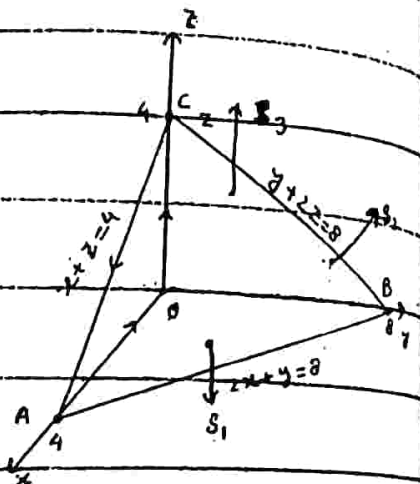
$$\oint_C \vec{F} \cdot d\vec{r} = \int_{AOCA} xz \, dx - y \, dy + x^2y \, dz$$

Let us we choose a particular

path the straight line segments

from $(4,0,0)$ to $(0,0,0)$ to $(0,0,4)$ to $(4,0,0)$ then

$$\begin{aligned}
 \oint_C \vec{F} \cdot d\vec{r} &= \int_{x=4}^0 0 \, dx + \int_{z=0}^4 0 \, dz + \int_{y=0}^4 (4x^2) \, dy \\
 &= 0 + 0 + 2x^2 \Big|_0^4 - \frac{x^3}{3} \Big|_0^4 \\
 &= 32 - \frac{64}{3} \\
 &= \frac{32}{3} \rightarrow \textcircled{2}
 \end{aligned}$$



New

$$\vec{F} = xz \hat{i} - y \hat{j} + x^2 y \hat{k}$$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz & -y & x^2 y \end{vmatrix}$$

$$= x^2 \hat{i} + (x - 2xy) \hat{j} + (0) \hat{k}$$

$$\nabla \times \vec{F} = x^2 \hat{i} + (x - 2xy) \hat{j}$$

$$\text{As } \iint_S (\nabla \times \vec{F}) \cdot \vec{n} \, ds = \iint_S [x^2 \hat{i} + x(1-2y) \hat{j}] \cdot \vec{n} \, ds \quad \text{--- (2)}$$

For S_1 :

$$\text{As } \vec{n} = -\hat{k}, \quad z=0, \quad 0 \leq x \leq 4; \quad 0 \leq y \leq 8$$

The equation (2) becomes

$$\iint_{S_1} (\nabla \times \vec{F}) \cdot \vec{n} \, ds_1 = \int_{x=0}^4 \int_{y=0}^8 0 \, dy \, dx = 0$$

For S_2 :

$$\text{As } \vec{n} = -\hat{i}, \quad x=0, \quad 0 \leq y \leq 8; \quad 0 \leq z \leq 4$$

The equation (2) becomes

$$\iint_{S_2} (\nabla \times \vec{F}) \cdot \vec{n} \, ds_2 = \int_{y=0}^8 \int_{z=0}^4 0 \, dz \, dy = 0$$

For S_3 :

$$\text{As } 2x + y + 2z = 8$$

$$\text{Let } \phi(x, y, z) = 2x + y + 2z - 8$$

$$\begin{aligned} \nabla \phi &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (2x + y + 2z - 8) \\ &= 2\hat{i} + \hat{j} + 2\hat{k} \end{aligned}$$

$$|\nabla \phi| = \sqrt{4+1+4} = 3$$

$$\begin{aligned} \vec{n} &= \frac{\nabla \phi}{|\nabla \phi|} \\ &= \frac{2\hat{i} + \hat{j} + 2\hat{k}}{3} \end{aligned}$$

$$|\vec{n} \cdot \hat{k}| = \frac{2}{3}$$

Now, $\iint_{S_3} (\nabla \times \vec{F}) \cdot \vec{n} \, dS_3 = \iint_R (\nabla \times \vec{F}) \cdot \vec{n} \frac{dx \, dy}{|\vec{h} \cdot \vec{n}|}$

$$\iint_{S_3} (\nabla \times \vec{F}) \cdot \vec{n} \, dS_3 = \iint_R [2x^2 \hat{i} + x(1-2y) \hat{j}] \cdot \left[\frac{2\hat{i} + \hat{j} + 2\hat{k}}{3} \right] \cdot \frac{dx \, dy}{3}$$

$$= \frac{1}{2} \iint_R [2x^2 + x(1-2y)] \, dx \, dy \quad \text{--- (3)}$$

$$\therefore 2x + y + 2z = 0$$

For xy -plane, $z=0$

$$2x + y = 0$$

For x -intercept, $y=0$

$$x=0$$

$$\text{As, } y = 8 - 2x$$

The Equation (3) becomes

$$\iint_{S_3} (\nabla \times \vec{F}) \cdot \vec{n} \, dS_3 = \frac{1}{2} \int_{x=0}^4 \int_{y=0}^{8-2x} [2x^2 + x(1-2y)] \, dy \, dx$$

$$= \frac{1}{2} \int_{x=0}^4 \left[2x^2 y + x(y - y^2) \right]_0^{8-2x} dx$$

$$= \frac{1}{2} \int_{x=0}^4 [2x^2(8-2x) + (8-2x - 64x + 4x^2 + 32x)x] dx$$

$$= \frac{1}{2} \int_{x=0}^4 [16x^2 - 4x^3 + 8x - 2x^2 - 64x + 4x^2 + 32x^2] dx$$

$$= \frac{1}{2} \int_{x=0}^4 [46x^2 - 8x^3 - 56x] dx$$

$$= \frac{1}{2} \left[\frac{46 \cdot x^3}{3} - 2x^4 - 28x^2 \right]_0^4$$

$$= \frac{1}{2} \left[\frac{46 \cdot 64}{3} - 2 \cdot 256 - 448 \right]$$

$$= \frac{1}{2} \left[\frac{2944 - 1536 - 1344}{3} \right]$$

$$= \frac{1}{2} \left[\frac{64}{3} \right]$$

$$= \frac{32}{3}$$

From equation (4)

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \vec{n} \, dS = \iint_{S_1} (\nabla \times \vec{F}) \cdot \vec{n} \, dS_1 + \iint_{S_2} (\nabla \times \vec{F}) \cdot \vec{n} \, dS_2 + \iint_{S_3} (\nabla \times \vec{F}) \cdot \vec{n} \, dS_3$$

Hence, The Stokes Theorem is verified.

63) Evaluate $\iint_S (\nabla \times \vec{A}) \cdot \vec{n} \, ds$ where $\vec{A} = (x^2 + y - 4)\hat{i} + 3xy\hat{j} + (2xz + z^2)\hat{k}$ and S is the surface of

(a) hemisphere $x^2 + y^2 + z^2 = 16$ above the xy -plane

(b) the paraboloid $z = 4 - (x^2 + y^2)$ above the xy -plane.

Sol.

Given that

$$\vec{A} = (x^2 + y - 4)\hat{i} + 3xy\hat{j} + (2xz + z^2)\hat{k}$$

$$\nabla \times \vec{A} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 + y - 4 & 3xy & 2xz + z^2 \end{vmatrix}$$

$$= \hat{i}(0) + \hat{j}(2z) + \hat{k}(3y - 1)$$

$$= 2z\hat{j} + (3y - 1)\hat{k}$$

(a) Now As $x^2 + y^2 + z^2 = a^2$ where $a = 4$

Let $\phi(x, y, z) = x^2 + y^2 + z^2 - a^2$

$$\nabla \phi = \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) (x^2 + y^2 + z^2 - a^2)$$

$$= 2x\hat{i} + 2y\hat{j} + 2z\hat{k}$$

$$|\nabla \phi| = \sqrt{4x^2 + 4y^2 + 4z^2} = 2a$$

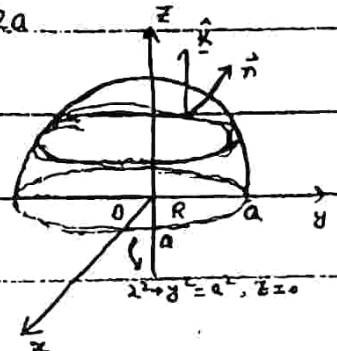
As,

$$\vec{n} = \frac{\nabla \phi}{|\nabla \phi|}$$

$$= \frac{2x\hat{i} + 2y\hat{j} + 2z\hat{k}}{2a}$$

$$\vec{n} = \frac{x}{a}\hat{i} + \frac{y}{a}\hat{j} + \frac{z}{a}\hat{k}$$

$$|\vec{n} \cdot \hat{k}| = \frac{z}{a}$$



Now $(\nabla \times \vec{A}) \cdot \vec{n} = (2z\hat{j} + (3y - 1)\hat{k}) \cdot \left(\frac{x}{a}\hat{i} + \frac{y}{a}\hat{j} + \frac{z}{a}\hat{k} \right)$

$$= \frac{2yz}{a} + \frac{(3y - 1)z}{a}$$

$$= \frac{(5y - 1)z}{a}$$

Now
$$\iint_S (\vec{\nabla} \times \vec{A}) \cdot \vec{n} \, ds = \iint_R (\vec{\nabla} \times \vec{A}) \cdot \vec{n} \frac{dx \, dy}{|\vec{n} \cdot \hat{k}|}$$

$$= \iint_R \frac{(5y-1) \cdot x}{x} \frac{dx \, dy}{x/a}$$

$$= \iint_R (5y-1) \, dx \, dy \rightarrow (1)$$

As $x^2 + y^2 + z^2 = a^2$

For xy -plane, $z=0$

$$x^2 + y^2 = a^2$$

For x -intercept $y=0$

$$x = \pm a$$

As $y = \pm \sqrt{a^2 - x^2}$

As x varies from $-a$ to a

y varies from $-\sqrt{a^2 - x^2}$ to $\sqrt{a^2 - x^2}$

The equation (1) becomes

$$\begin{aligned} \iint_S (\vec{\nabla} \times \vec{A}) \cdot \vec{n} \, ds &= \int_{x=-a}^a \int_{y=-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} (5y-1) \, dy \, dx \\ &= \int_{x=-a}^a \left(\frac{5y^2}{2} - y \right) \bigg|_{y=-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} dx \end{aligned}$$

$$= \int_{x=-a}^a -2\sqrt{a^2-x^2} \, dx \rightarrow (2)$$

Let $x = a \sin \theta$

$$dx = a \cos \theta \, d\theta$$

When $x = -a$ then $\theta = -\pi/2$

When $x = a$ then $\theta = \pi/2$

The equation (2) becomes

$$\iint_S (\vec{\nabla} \times \vec{A}) \cdot \vec{n} \, ds = -2 \int_{\theta=-\pi/2}^{\pi/2} a \cos \theta \cdot a \cos \theta \, d\theta$$

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$$\begin{aligned}
 \iint_S (\vec{\nabla} \times \vec{A}) \cdot \vec{n} \, ds &= -2a^2 \cdot \int_{\theta=-\pi/2}^{\pi/2} \cos^2 \theta \, d\theta \\
 &= -2a^2 \cdot \left(\theta + \frac{\sin 2\theta}{2} \right) \Big|_{\theta=-\pi/2}^{\pi/2} \\
 &= -\frac{2a^2}{2} \cdot (\pi/2 + \pi/2) \\
 &= -\pi a^2
 \end{aligned}$$

$$\therefore a = 4$$

$$= -16\pi$$

(b) As, $z = 4 - (x^2 + y^2)$

Let $\phi(x, y, z) = x^2 + y^2 + z - 4$

$$\begin{aligned}
 \vec{\nabla} \phi &= \left(\frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k} \right) \cdot (x^2 + y^2 + z - 4) \\
 &= 2x \hat{i} + 2y \hat{j} + \hat{k}
 \end{aligned}$$

$$|\vec{\nabla} \phi| = \sqrt{4+4+1} = 3$$

As, $\vec{n} = \frac{\vec{\nabla} \phi}{|\vec{\nabla} \phi|}$

$$= \frac{2x \hat{i} + 2y \hat{j} + \hat{k}}{3}$$

$$|\vec{n} \cdot \hat{k}| = \frac{1}{3}$$

So, $(\vec{\nabla} \times \vec{A}) \cdot \vec{n} = \frac{(2z \hat{j} + (3y-1) \hat{k}) \cdot (2x \hat{i} + 2y \hat{j} + \hat{k})}{3}$

$$= \frac{4yz + 3y - 1}{3}$$

Now $\iint_S (\vec{\nabla} \times \vec{A}) \cdot \vec{n} \, ds = \iint_R \frac{(\vec{\nabla} \times \vec{A}) \cdot \vec{n}}{|\vec{n} \cdot \hat{k}|} \, dx \, dy$

$$\begin{aligned}
 &= \iint_R \frac{4yz + 3y - 1}{3} \cdot \frac{1}{1/3} \, dx \, dy \\
 &= \iint_R (4yz + 3y - 1) \, dx \, dy \\
 &= \iint_R (4y(4 - x^2 - y^2) + 3y - 1) \, dx \, dy \\
 &= \iint_R (16y - 4y(x^2 + y^2) + 3y - 1) \, dx \, dy \\
 &= \iint_R (19y - 4y(x^2 + y^2) - 1) \, dx \, dy \quad \rightarrow \textcircled{2} \\
 &= \iint_R (19y - 4yx^2 - 4y^3 - 1) \, dx \, dy
 \end{aligned}$$

As $z = 4 - (x^2 + y^2)$

For xy -plane $z = 0$

$$x^2 + y^2 = 4$$

For x -intercept $y = 0$

$$x = \pm 2$$

As $y = \pm \sqrt{4 - x^2}$

As x varies from -2 to 2

y varies from $-\sqrt{4 - x^2}$ to $\sqrt{4 - x^2}$

The equation (1) becomes

$$\begin{aligned} \iint_S (\vec{\nabla} \times \vec{A}) \cdot \vec{n} \, ds &= \int_{x=-2}^2 \int_{y=-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (19y - 4x^2y - 4y^3 - 1) \, dy \, dx \\ &= \int_{x=-2}^2 \left(19 \frac{y^2}{2} - 2x^2 y^2 - y^4 - y \right) \bigg|_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} dx \\ &= \int_{x=-2}^2 -2\sqrt{4-x^2} \, dx \quad \text{--- (2)} \end{aligned}$$

Let $x = 2 \sin \theta$

$$dx = 2 \cos \theta \, d\theta$$

when $x = 2$ then $\theta = +\pi/2$

when $x = -2$ then $\theta = -\pi/2$

The equation (2) becomes

$$\begin{aligned} \iint_S (\vec{\nabla} \times \vec{A}) \cdot \vec{n} \, ds &= -2 \int_{\theta=-\pi/2}^{\pi/2} 2 \cos \theta \cdot 2 \cos \theta \, d\theta \\ &= -8 \int_{\theta=-\pi/2}^{\pi/2} \frac{1 + \cos 2\theta}{2} \, d\theta \\ &= -4 \left(\theta + \frac{\sin 2\theta}{2} \right) \bigg|_{-\pi/2}^{\pi/2} \\ &= -4 \left(\pi/2 + \pi/2 \right) \\ &= -4\pi \end{aligned}$$

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Q6) If $\vec{A} = 2yz\hat{i} - (x+3y-2)\hat{j} + (x^2+z)\hat{k}$, evaluate $\iint_S (\nabla \times \vec{A}) \cdot \vec{n} ds$ over the surface of intersection of the cylinders $x^2+z^2=a^2$ and $x^2+y^2=a^2$ which is included in the first octant.

Sol:-

Given that

$$\vec{A} = 2yz\hat{i} - (x+3y-2)\hat{j} + (x^2+z)\hat{k}$$

As,

$$\iint_S (\nabla \times \vec{A}) \cdot \vec{n} ds = \oint_C \vec{A} \cdot d\vec{r}$$

$$= \oint_{ABCD} 2yz dx - (x+3y-2) dy + (x^2+z) dz \quad \rightarrow (1)$$

For AB

$$\text{As, } z=0 \text{ so, } dz=0$$

$$\int_{AB} = \int_{AB} -(x+3y-2) dy = \int_{y=0}^a -(\sqrt{a^2-y^2} + 3y-2) dy \quad \rightarrow (2)$$

$$\text{Let } y = a \sin \theta$$

$$dy = a \cos \theta d\theta$$

$$\text{As, } 0 \leq \theta \leq \pi/2$$

The equation (2) becomes

$$\int_{AB} = \int_{\theta=0}^{\pi/2} -(a \cos \theta + 3a \sin \theta - 2) a \cos \theta d\theta$$

$$= \int_{\theta=0}^{\pi/2} (-a^2 \cos^2 \theta - 3a^2 \sin \theta \cos \theta + 2a \cos \theta) d\theta$$

$$= \left(-\frac{a^2}{2} (\cos \theta + \sin 2\theta) - \frac{3}{2} a^2 \sin^2 \theta + 2a \sin \theta \right) \Big|_{\theta=0}^{\pi/2}$$

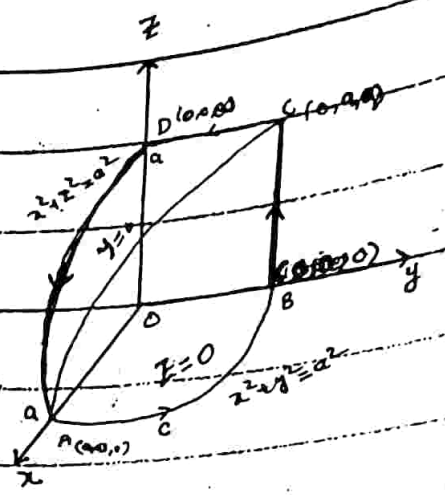
$$= \frac{a^2}{2} (\pi/2) - \frac{3}{2} a^2 + 2a$$

$$= \frac{\pi a^2}{4} - \frac{3}{2} a^2 + 2a$$

For BC

$$\text{As, } x=0, y=a, dx=dy=0$$

$$\int_{BC} = \int_{z=0}^a z dz = \frac{a^2}{2}$$



For CD: $x = a, y = 0, z = 0, dx = dy = 0$

$$\begin{aligned} \int_{CD} &= \int_{y=0}^0 -(3y-2) = \int_{y=0}^0 (3y-2) = \left[\frac{3}{2} y^2 - 2y \right]_0^0 \\ &= \frac{3}{2} a^2 - 2a \end{aligned}$$

For DA:

$$\text{As, } y=0, dy=0$$

$$\begin{aligned} \int_{DA} &= \int_{z=a}^0 (x^2 + z) \\ &= \int_{z=a}^0 (a^2 - z^2 + z) dz \\ &= \int_{z=a}^0 (a^2 - z^2 + z) dz \\ &= \left[a^2 z - \frac{z^3}{3} + \frac{z^2}{2} \right]_a^0 \\ &= -a^3 + \frac{a^3}{3} - \frac{a^2}{2} \\ &= -\frac{2a^3}{3} - \frac{a^2}{2} \end{aligned}$$

The Equation (4) Becomes

$$\begin{aligned} \iint_S (\nabla \times \vec{A}) \cdot \vec{n} \, ds &= \int_{AB} + \int_{BC} + \int_{CD} + \int_{DA} \\ &= -\frac{\pi a^2}{4} - \frac{3a^2}{2} + \frac{a^2}{2} + \frac{3a^2}{2} - \frac{2a^3}{3} - \frac{a^2}{2} \\ &= -\frac{\pi a^2}{4} - \frac{2a^3}{3} \\ &= -\frac{a^2}{12} (3\pi + 8a) \end{aligned}$$

* * *

(67) A vector $\vec{B}$ is always normal to a given closed surface S . Show that $\iiint_V \text{curl } \vec{B} \, dV = 0$ where V is the Region bounded by S

Sol:-

We know by CURL Theorem

$$\iiint_V \nabla \times \vec{B} \, dV = \iint_S \vec{n} \times \vec{B} \, ds$$

Let $\vec{B} = B\vec{n}$

($\therefore$ Normal to the given closed surface)

Then

$$\begin{aligned}\iiint_V \vec{\nabla} \times \vec{B} \, dV &= \iint_S \vec{n} \times B\vec{n} \, dS \\ &= \iint_S B(\vec{n} \times \vec{n}) \, dS \\ &= 0\end{aligned}$$

(68) If $\oint_C \vec{E} \cdot d\vec{r} = -\frac{1}{c} \frac{\partial}{\partial t} \iint_S \vec{H} \cdot d\vec{s}$ where S is any surface bounded by a curve "C" show that

$$\vec{\nabla} \times \vec{E} = -\frac{1}{c} \frac{\partial \vec{H}}{\partial t}$$

Sol:-

Given That

$$\oint_C \vec{E} \cdot d\vec{r} = -\frac{1}{c} \frac{\partial}{\partial t} \iint_S \vec{H} \cdot d\vec{s}$$

$$\oint_C \vec{E} \cdot d\vec{r} = \iint_S -\frac{1}{c} \frac{\partial \vec{H}}{\partial t} \cdot d\vec{s} \quad \rightarrow (1)$$

We know By Stokes Theorem

$$\oint_C \vec{E} \cdot d\vec{r} = \iint_S (\vec{\nabla} \times \vec{E}) \cdot d\vec{s} \quad \rightarrow (2)$$

By Comparing (1) and (2)

$$\begin{aligned}\iint_S -\frac{1}{c} \frac{\partial \vec{H}}{\partial t} \cdot d\vec{s} &= \iint_S (\vec{\nabla} \times \vec{E}) \cdot d\vec{s} \\ \vec{\nabla} \times \vec{E} &= -\frac{1}{c} \frac{\partial \vec{H}}{\partial t}\end{aligned}$$

(69) Prove $\oint_C \phi \, d\vec{r} = \iint_S d\vec{s} \times \nabla \phi$

Sol:-

Let $\vec{A} = \phi \vec{C}$ where $\vec{C}$ is constant vector

By Stokes Theorem

$$\oint_C \vec{A} \cdot d\vec{r} = \iint_S (\vec{\nabla} \times \vec{A}) \cdot d\vec{s}$$

$$\oint_C \phi \vec{c} \cdot d\vec{r} = \iint_S [\nabla \times (\phi \vec{c})] \cdot d\vec{s}$$

$$\oint_C \vec{c} \cdot \phi d\vec{r} = \iint_S \nabla \phi \times \vec{c} \cdot d\vec{s}$$

$$\oint_C \vec{c} \cdot \phi d\vec{r} = \iint_S \vec{c} \cdot d\vec{s} \times \nabla \phi$$

$$\vec{c} \cdot \oint_C \phi d\vec{r} = \vec{c} \cdot \iint_S d\vec{s} \times \nabla \phi$$

$$\oint_C \phi d\vec{r} = \iint_S d\vec{s} \times \nabla \phi$$

$$\begin{aligned} \oint_C \phi d\vec{r} &= \oint_C \phi \vec{c} \cdot d\vec{r} \\ &= \oint_C \phi \vec{c} \cdot d\vec{r} \\ &= \oint_C \phi \vec{c} \cdot d\vec{r} \end{aligned}$$

(71) Prove $\iiint_V \nabla \phi \cdot \vec{A} \, dv = \iint_S \phi \vec{A} \cdot \vec{n} \, ds - \iiint_V \phi \nabla \cdot \vec{A} \, dv$

Soln

Let $\vec{B} = \phi \vec{A}$

By Divergence Theorem

$$\iiint_V \nabla \cdot \vec{B} \, dv = \iint_S \vec{B} \cdot \vec{n} \, ds$$

$$\iiint_V \nabla \cdot (\phi \vec{A}) \, dv = \iint_S \phi \vec{A} \cdot \vec{n} \, ds$$

$$\iiint_V (\nabla \phi \cdot \vec{A} + \phi \nabla \cdot \vec{A}) \, dv = \iint_S \phi \vec{A} \cdot \vec{n} \, ds$$

$$\iiint_V \nabla \phi \cdot \vec{A} \, dv + \iiint_V \phi \nabla \cdot \vec{A} \, dv = \iint_S \phi \vec{A} \cdot \vec{n} \, ds$$

$$\iiint_V \nabla \phi \cdot \vec{A} \, dv = \iint_S \phi \vec{A} \cdot \vec{n} \, ds - \iiint_V \phi \nabla \cdot \vec{A} \, dv$$

(72) Let $\vec{r}$ be the position vector of any point relative to an origin O. Suppose ϕ have continuous derivatives of order two at least and let S be a closed surface bounding a volume V. Denote ϕ at O by ϕ_0 . Show that

$$\iint_S [\frac{1}{r} \nabla \phi - \phi \nabla (\frac{1}{r})] \cdot d\vec{s} = \iiint_V \nabla^2 \phi \, dv + \alpha$$

where $\alpha = 0$ or $\alpha = 4\pi\phi_0$ according as O is outside or inside S.

OR

Prove that the Green's Third identity

OR

Prove that the Green's Formula.

Proof: Case 1: when origin lies outside S then

We know by Green's Second identity

$$\iiint_V (\phi \nabla^2 \psi - \psi \nabla^2 \phi) dv = \iint_S (\phi \nabla \psi - \psi \nabla \phi) \cdot \vec{n} dS$$

Let $\psi = 1/r$ which is harmonic function.

$$\text{Then } \iiint_V (\phi \nabla^2 (1/r) - 1/r \nabla^2 \phi) dv = \iint_S (\phi \nabla (1/r) - 1/r \nabla \phi) \cdot \vec{n} dS$$

We know $\nabla^2 (1/r) = 0$ if $r \neq 0$

So,

$$\iiint_V (0 - 1/r \nabla^2 \phi) = \iint_S (\phi \nabla (1/r) - 1/r \nabla \phi) \cdot \vec{n} dS$$

So,

$$-\iiint_V 1/r \nabla^2 \phi = + \iint_S (\phi \nabla (1/r) - 1/r \nabla \phi) \cdot \vec{n} dS$$

$$\iint_S [1/r \nabla \phi - \phi \nabla (1/r)] \cdot \vec{n} dS = \iiint_V 1/r \nabla^2 \phi + \alpha$$

where $\alpha = 0$

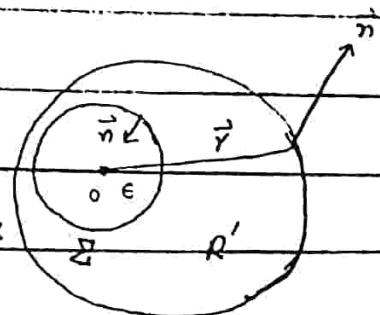
Case 2:-

If origin "o" lies inside S .

Surround "o" by a small sphere

Σ with center at "o" and radius

" ϵ "



Let V be the Region bounded by S and Σ

Then

$$\iint_{S+\Sigma} (\phi \nabla (1/r) - 1/r \nabla \phi) \cdot \vec{n} dS = - \iiint_V 1/r \nabla^2 \phi dv$$

$$\iint_{S+\Sigma} (1/r \nabla \phi - \phi \nabla (1/r)) \cdot \vec{n} dS = \iiint_V 1/r \nabla^2 \phi dv$$

$$\iint_S \left(\frac{1}{r} \nabla \phi - \phi \nabla \left(\frac{1}{r} \right) \right) ds + \iint_S \left(\frac{1}{r} \bar{\nabla} \phi - \phi \bar{\nabla} \left(\frac{1}{r} \right) \right) ds = \iiint_V \frac{1}{r} \nabla^2 \phi \, dv \quad \text{c. 1}$$

Now on Σ :

$$r = e \quad \text{and} \quad \vec{n} = -\hat{r} = -\frac{\vec{r}}{r} = -\frac{\vec{r}}{e} \quad \text{c. 1}$$

$$\text{As, } \vec{\nabla} \left(\frac{1}{r} \right) = \frac{\partial}{\partial n} \left(\frac{1}{r} \right) = -\frac{\partial}{\partial r} \left(\frac{1}{r} \right) = -\left(-\frac{1}{r^2} \right) = \frac{1}{e^2} \quad \left| \frac{\partial}{\partial n} \left(\frac{1}{r} \right) \right|$$

As,

$$\begin{aligned} \iint_{\Sigma} \left(\frac{1}{r} \bar{\nabla} \phi - \phi \bar{\nabla} \left(\frac{1}{r} \right) \right) ds &= \iint_{\Sigma} \left(\frac{1}{e} \bar{\nabla} \phi - \phi \cdot \frac{1}{e^2} \right) ds \\ &= \left(\frac{1}{e} \bar{\nabla} \phi - \phi \frac{1}{e^2} \right) \iint_{\Sigma} ds \\ &= \left(\frac{1}{e} \bar{\nabla} \phi - \phi \frac{1}{e^2} \right) 4\pi e^2 \\ &\approx -4\pi \phi_0 \quad \text{if } e \gg 0 \end{aligned}$$

where ϕ_0 is the value of ϕ at "O".

So,

$$\iint_S \left(\frac{1}{r} \bar{\nabla} \phi - \phi \bar{\nabla} \left(\frac{1}{r} \right) \right) ds - 4\pi \phi_0 = \iiint_V \frac{1}{r} \nabla^2 \phi \, dv$$

$$\iint_S \left(\frac{1}{r} \bar{\nabla} \phi - \phi \bar{\nabla} \left(\frac{1}{r} \right) \right) ds = \iiint_V \left(\frac{1}{r} \nabla^2 \phi \right) dv + 4\pi \phi_0$$

$$\iint_S \left(\frac{1}{r} \bar{\nabla} \phi - \phi \bar{\nabla} \left(\frac{1}{r} \right) \right) ds = \iiint_V \left(\frac{1}{r} \nabla^2 \phi \right) dv + \alpha$$

where $\alpha = 4\pi \phi_0$